# Detection, Adaptation, and Watermark Persistence
# A Pedagogical Research Report

**Abstract**

This report studies how AI-content detectors generalize, how generators learn to evade them, and how text and image watermarks survive removal attempts. Short method explanations introduce the mathematics needed to interpret each experiment. The latest results cover a fresh 128-question TextSeal 2 comparison, a support-pruning pilot, a 4,096-prefix false-positive audit, and the post-hoc reanalysis of image preservation.

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# 1 Research questions and reading guide

This project asks how AI-content detectors generalize, what happens when generators adapt to them, and how long deliberate watermark signals survive. Detector scores measure statistical evidence. A

watermark differs from a content classifier because a provider deliberately inserts the signal that its detector seeks.

Part I covers supervised text and image detection. Part II introduces GRPO and diffusion policy optimization alongside the evasion experiments. Part III explains watermark testing and compares text and image attacks. Part IV follows the newer experiments on image meta-distillation and the probabilities exposed by a text API. Each part introduces the mathematics needed for its results.

Read each result alongside the attacker's access, the held-out cohort, baseline detection, and the preservation rule. For example, fewer successful rewrites can mean either that the watermark survived or that the new answers failed the content checks. The tables report those outcomes separately.

**Notation used in the report.** Each mathematical section states its own objects and dimensions. The sets $\mathbb{R}$, $\mathbb{N}_{>0}$ and $\mathbb{N}_0$ contain real numbers, positive integers and nonnegative integers. For $d, m, n \in \mathbb{N}_{>0}$, a scalar has one value, $a \in \mathbb{R}^d$ is a vector with $d$ coordinates, and $A \in \mathbb{R}^{m\times n}$ is an $m$-row, $n$-column matrix. For a finite set $\mathcal{V}$ of size $V \in \mathbb{N}_{>0}$, its probability simplex is

$$\Delta(\mathcal{V}) = \left\{ p = (p_v)_{v\in\mathcal{V}} \in [0,1]^V : \sum_{v\in\mathcal{V}} p_v = 1 \right\}.$$

The set $\mathcal{V}^*$ contains all finite token sequences, including the empty sequence. The notation $\mathbf{1}\{E\} \in \{0,1\}$ indicates whether statement $E$ holds. A superscript $\top$ denotes transpose, $\|\cdot\|_2$ is the Euclidean norm, and $\mathbb{E}$ denotes expectation under the stated distribution. For parameters $\theta \in \mathbb{R}^{d_\theta}$ with $d_\theta \in \mathbb{N}_{>0}$, the gradient of a differentiable scalar function $f : \mathbb{R}^{d_\theta} \to \mathbb{R}$ has dimension $\nabla_\theta f \in \mathbb{R}^{d_\theta}$, with one entry per parameter. Logarithms are natural. For discrete distributions $p, q \in \Delta(\mathcal{V})$, their scalar Kullback–Leibler divergence is

$$\mathrm{KL}(p\|q) = \sum_{v:p_v>0} p_v \log(p_v/q_v) \in [0,\infty],$$

with infinite divergence if $q_v = 0 < p_v$. The continuous analogue integrates the density-weighted log ratio. Numerical parameters in displayed examples illustrate a mechanism; experiment tables give the measured outcomes.

# Part I
# Supervised detection and generalization

## 2 Text detection: what generalizes across datasets?

**Question.** Before adapting a generator, the project first asks which supervised detectors transfer beyond their fitting distribution. This part requires only the supervised-learning distinction between training, validation and test sets. Its results are independent of the subsequent reinforcement-learning and watermark experiments.

**Models and evidence.** The retained MAGE notebook compares four TF-IDF classifiers (logistic regression, random forest, XGBoost and LightGBM) with six QLoRA classifiers: ModernBERT-base, ModernBERT-large, Qwen3.5-4B-Base, Gemma-4-E4B, Nanbeige4.2-3B and LFM2.5-2.6B. Text labels in this notebook are machine $= 0$ and human $= 1$. The benchmark reports accuracy, macro F1, ROC-AUC and PR-AUC. Macro F1 gives equal weight to the two class-specific F1 values; it is not automatically the F1 of the AI class alone.

| Model | MAGE F1 | Pile F1 | Peer-review F1 |
|---|---|---|---|
| TF-IDF logistic regression | .6588 | .6670 | .6236 |
| TF-IDF random forest | .6687 | .6321 | .6105 |
| TF-IDF XGBoost | .6836 | .6109 | .6259 |
| TF-IDF LightGBM | .6994 | .6228 | .6757 |
| ModernBERT-base | .7908 | .7282 | .6249 |
| ModernBERT-large | .8153 | .7473 | .7700 |
| Qwen3.5-4B-Base | .9236 | .8310 | .8093 |
| Gemma-4-E4B | .9075 | .8006 | .9499 |
| Nanbeige4.2-3B | .7004 | .7272 | .6223 |
| LFM2.5-2.6B | .9061 | .7633 | .9154 |

Table 1: Macro F1 from the retained text notebook, job 1145971. Each column is a different evaluation distribution, not an independent retraining of every model.

**Reading the result.** Qwen has the highest observed MAGE and Pile macro F1, while Gemma has the highest peer-review macro F1. A model ranking on one dataset therefore does not define a universal detector ranking. An encoder can be cheaper to use than a decoder classifier, but this experiment does not provide a matched-compute efficiency frontier.

The peer-review evaluation contains 2,000 items: 1,000 human reviews and 200 generated reviews from each of GPT-4o, Gemini-1.5-Pro, Claude-Sonnet-3.5, Llama-3.1-70B and Qwen-2.5-72B. Its text extraction omits metadata fields such as review ratings and identifiers. This reduces an obvious metadata shortcut, but the evaluation changes both domain and generator. It cannot identify a purely temporal generalization effect. The executed MAGE dataset did not expose the additional MAGE-specific OOD splits named in the notebook introduction; those absent split evaluations must not be reported as completed.

**Why this matters later.** The defender-comparison workflow selects the strongest retained non-Qwen decoder and the strongest encoder from the MAGE comparison, then warms them against Qwen-generated continuations. Because selection uses the MAGE test ranking, that ranking is a selection statistic for the later study, not a second untouched confirmation. The CNN/DailyMail prompt evaluation remains a distinct experiment with its own split and overlap audit.

**Limits.** These are single-run observations without uncertainty intervals. High classifier scores are not calibrated authorship probabilities. Token limits, class order and dataset-specific style can alter apparent performance. Neither these scores nor later evasion results certify the truth of a document's content.

## 3 Image detection: dataset shift and a failed intervention

**Question.** Can a classifier trained on a broad collection of real and generated images recognize images from a different acquisition pipeline? This part is independent of DDPO and watermarking: it concerns supervised classifiers and their generalization.

The image notebook starts from ArtiFact's 2,440,031 available images, takes a seeded stratified 15% sample, and reserves a stratified 20% holdout from that sample. This is an image-level holdout, not a guarantee of unseen-generator evaluation. Five models are compared: EfficientNet-B4, a CLIP classifier, ConvNeXt V2 Tiny, DINOv2 ViT-base and ResNet-50. CIFAKE and Shoes serve as external evaluations. The CIFAKE input resolution differs sharply from the ArtiFact pipeline, making resolution and processing part of the shift.

| Model | ArtiFact F1 | CIFAKE F1 | Shoes F1 |
|---|---|---|---|
| EfficientNet-B4 | .9346 | .3333 | .6700 |
| CLIP classifier | .8039 | .3377 | .6732 |
| ConvNeXt V2 Tiny | .3811 | .3333 | .3834 |
| DINOv2 ViT-base | .6245 | .4564 | .3233 |
| ResNet-50 | .7562 | .3786 | .5398 |

Table 2: Retained notebook execution, job 1153220. Macro F1 exposes single-class collapse that accuracy can obscure.

For example, a classifier predicting one class on a balanced dataset has 50% accuracy but macro F1 of one third: the predicted class has precision $1/2$ and recall one, hence F1 $2/3$, while the other class has F1 zero. Averaging yields $1/3$. Thus the CIFAKE rows near 0.3333 are consistent with unhelpful single-class predictions, not moderately successful discrimination.

**The intervention and its rejection.** The project retrains ConvNeXt on balanced ArtiFact data with downsampling/upsampling, JPEG recompression, blur, color changes and flips. The prespecified exploratory gate requires macro-F1 improvement on both external datasets and no more than 0.03 loss on ArtiFact. Its macro-F1 changes are zero on all three evaluations, so the candidate is rejected. An increase in ranking performance without a corresponding thresholded decision improvement is not a pass of that gate. The retained Shoes AUROC changes substantially, but the operational macro-F1 criterion still fails.

The notebook also retains an earlier historical motivation describing 99.9% ArtiFact accuracy. That number belongs to an earlier execution. It must not replace the current ConvNeXt result of 61.58% accuracy and 0.3811 macro F1. Preserving both with explicit provenance is necessary to avoid an internally contradictory report.

**Interpretation.** The transfer gap is compatible with shortcuts involving resolution, compression or content sources, but the benchmark alone does not identify their causal contributions. External results informed the intervention design, so even a successful exploratory intervention would need untouched confirmation. These limitations motivate the later independent official-DDA evaluation rather than a claim that a high ArtiFact holdout score establishes robust image forensics.

# Part II
# Adaptive generation and defender recovery

## 4 GRPO for text generation

Group Relative Policy Optimization (GRPO) (Shao et al., 2024) learns from scores on sampled answers. A language model is a policy because it chooses a distribution over the next token. A detector scores a completed answer, and the update makes the tokens of better answers more likely. This supplies a gradient through the language model's log probabilities even when the detector itself has no gradient. Group comparison supplies the baseline without training a separate value network.

**Tokens, policies and rewards.** Let $\mathcal{V}$ be a finite vocabulary of size $V = |\mathcal{V}| \in \mathbb{N}_{>0}$, and let $c \in \mathcal{V}^*$ be a prompt, where $\mathcal{V}^*$ is the set of finite token sequences. The trainable parameters are $\theta \in \mathbb{R}^{d_\theta}$, with $d_\theta \in \mathbb{N}_{>0}$. At any prompt and generated prefix, $\pi_\theta(\cdot \mid c, y_{<t}) \in \Delta(\mathcal{V})$ is a length-$V$ probability

vector. Here $y_{<t}$ contains the tokens before position $t$, and $\Delta(\mathcal{V})$ is the probability simplex defined in the reading guide.

Freeze a rollout policy $\pi_{\text{old}}$, whose conditional distributions also belong to $\Delta(\mathcal{V})$, and sample $G \in \{2, 3, ...\}$ answers. Answer $i \in \{1, \dots, G\}$ has length $L_i \in \mathbb{N}_{>0}$ and tokens $y_i = (y_{i1}, \dots, y_{iL_i}) \in \mathcal{V}^{L_i}$. A fixed scoring rule returns a scalar reward $r_i \in \mathbb{R}$, combining evasion and content utility in this project. Define the mean reward $\bar{r} \in \mathbb{R}$ and relative advantage $A_i \in \mathbb{R}$ by

$$\bar{r} = \frac{1}{G}\sum_{j=1}^{G} r_j, \qquad A_i = \frac{r_i - \bar{r}}{s_r + \varepsilon}.$$

The scalar $s_r \in [0, \infty)$ is the group's reward standard deviation under the trainer's reduction convention, and $\varepsilon \in (0, \infty)$ stabilizes division. With rewards $(0.2, 0.4, 0.6, 0.8) \in \mathbb{R}^4$, the mean is $0.5$ and the last two answers receive positive feedback. Identical rewards give zero advantages. Relative ranking can favor an answer even when every answer is poor.

**Why probability ratios and clipping enter.** The answers were sampled before the update. At each sampled response position $t \in \{1, \dots, L_i\}$, compare its current and rollout probabilities using the scalar ratio

$$u_{it}(\theta) = \frac{\pi_\theta(y_{it} \mid c, y_{i,<t})}{\pi_{\text{old}}(y_{it} \mid c, y_{i,<t})} \in [0, \infty).$$

The denominator is positive for a sampled token. A response-averaged clipped objective is the scalar

$$\mathcal{J}_{\text{clip}}(\theta) = \frac{1}{G}\sum_{i=1}^{G}\frac{1}{L_i}\sum_{t=1}^{L_i} \min\left\{u_{it}(\theta)A_i,\ \text{clip}\left(u_{it}(\theta), 1-\epsilon, 1+\epsilon\right)A_i\right\}.$$

Here $\epsilon \in (0, 1)$ is the clipping width, distinct from the numerical stabilizer $\varepsilon$. For real scalars $a, l, u \in \mathbb{R}$ with $l \le u$, $\text{clip}(a, l, u)$ restricts $a$ to $[l, u]$. Maximizing this objective initially increases the probabilities of above-average answers and decreases those of below-average answers. For $A_i = 1$, $u_{it} = 1.5$ and $\epsilon = 0.2$, the contribution is capped at $1.2$. With $A_i = -1$ and $u_{it} = 0.5$, it is $-0.8$. Clipping limits the incentive for further change in the rewarded direction on these sampled tokens.

A reference policy $\pi_{\text{ref}}(\cdot \mid c, y_{<t}) \in \Delta(\mathcal{V})$ can supply an additional scalar KL penalty, weighted by $\beta \in [0, \infty)$, to discourage distributional drift. The retained TRL runs use $\beta = 0$ and separate content rewards. Their exact response masks and loss reductions follow the saved trainer configuration. The manual fallback uses group-normalized REINFORCE, which updates sampled-token log probabilities with relative rewards. The project also implements Self-Play Fine-Tuning (SPIN) and Model-Agnostic Meta-Learning (MAML). SPIN uses preferred/rejected answer pairs; MAML trains an initialization using support data for adaptation and query data for the outer update. Both splits contribute to MAML training, so evaluation requires a separate test split.

# 5 Detection and adaptive text generation

All detectors expose an AI-oriented score: higher indicates stronger generated-content evidence. An ensemble averages seven component scores. Averaging does not calibrate them into posterior probabilities. Learned classifiers, likelihood statistics and handcrafted heuristics can respond differently to a distribution shift.

The retained Qwen3.5-9B-Base LoRA attackers optimize a mixture of detector evasion, semantic similarity and quality rewards. Three final checkpoints target Gemma, ModernBERT-large and the ensemble. Their training budgets differ, so final evasion rates are not a matched-budget comparison of defender architectures.

| Target | Updates | Evasion before defense | After defense |
|---|---|---|---|
| Gemma | 1,213 | 77.2% | 0.2% |
| ModernBERT-large | 1,487 | 96.8% | 1.0% |
| Seven-detector ensemble | 502 | 94.8% | 0.4% |

Table 3: One defender update, evaluated on identical saved attack texts: 500 generated and 500 human passages per branch, threshold 0.5.

Each defender update uses fresh fitting outputs while evaluation reuses the original saved texts. This isolates changes to the defender. It establishes recovery against a fixed attacker. It does not establish resistance to a second adaptive RL round. Single-seed results, unequal training budgets and diagnostic reuse limit generalization claims.

The RoBERTa diagnostic illustrates checkpoint instability: evasion on 100 prompts reaches 98% at step 500 and falls to 36% at step 600. Selecting a favorable checkpoint after viewing diagnostic results can overstate robustness or attack reliability. Confirmatory testing must use untouched prompts and a prespecified selection rule.

# 6 Adaptive image generation

The image attacker changes Stable Diffusion so its outputs score lower on a detector while remaining aligned with their prompts. Text GRPO assigns probabilities to tokens. Here the policy assigns densities to steps in the denoising process. The next two sections explain how that difference affects the update.

# 7 Diffusion as a sequence of actions

Diffusion training teaches a network to remove noise at different corruption levels. Work with one flattened latent $x_0 \in \mathbb{R}^d$, where $d = C_z H_z W_z$ and $C_z, H_z, W_z \in \mathbb{N}_{>0}$ are its channel, height and width dimensions. The equivalent tensor has shape $C_z \times H_z \times W_z$. A decoder $\mathcal{D} : \mathbb{R}^d \to \mathbb{R}^{H \times W \times 3}$ turns the final latent into an RGB image with $H, W \in \mathbb{N}_{>0}$. The derivation concerns one sample; a batch of $B_{\text{batch}} \in \mathbb{N}_{>0}$ latents has shape $B_{\text{batch}} \times C_z \times H_z \times W_z$.

Let $T \in \mathbb{N}_{>0}$ be the number of training noise levels, $t \in \{1, \dots, T\}$ a level, and $\bar{\alpha}_t \in (0, 1)$ the scalar cumulative signal retention. With noise $\varepsilon \sim \mathcal{N}(0_d, I_d)$ in $\mathbb{R}^d$, where $0_d \in \mathbb{R}^d$ and $I_d \in \mathbb{R}^{d \times d}$ are the zero vector and identity matrix, a noisy training latent is

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \varepsilon \in \mathbb{R}^d.$$

For text conditioning, let $c \in \mathcal{C} = \mathbb{R}^{L_c \times d_c}$ be the prompt-embedding matrix, where $L_c, d_c \in \mathbb{N}_{>0}$ count encoded positions and embedding coordinates. With the sampling and guidance settings fixed, the denoiser $\varepsilon_\theta : \mathbb{R}^d \times \{1, \dots, T\} \times \mathcal{C} \to \mathbb{R}^d$ predicts that noise. Its parameters are $\theta \in \mathbb{R}^{d_\theta}$ with $d_\theta \in \mathbb{N}_{>0}$. Mean squared noise error trains the network. During generation, a scheduler starts from noise and repeatedly uses the predictions to produce a less noisy latent.

For the stochastic Denoising Diffusion Implicit Model (DDIM) sampler used in the evasion experiments, let $s \in \{0, \dots, t-1\}$ be the next scheduled index. At fixed $(x_t, t, c)$, the next latent has Gaussian density

$$p_\theta(x_s \mid x_t, c) = \mathcal{N}(x_s; \mu_\theta, \sigma_{t \to s}^2 I_d).$$

The scheduler and denoiser determine the mean vector $\mu_\theta = \mu_\theta(x_t, t, c) \in \mathbb{R}^d$ and scalar standard deviation $\sigma_{t \to s} \in [0, \infty)$. For $\sigma_{t \to s} > 0$, the scalar log density of a saved successor $x_s \in \mathbb{R}^d$ is

$$\ell_\theta = \log p_\theta(x_s \mid x_t, c) = -\frac{\|x_s - \mu_\theta\|_2^2}{2\sigma_{t \to s}^2} - \frac{d}{2} \log(2\pi\sigma_{t \to s}^2) \in \mathbb{R}.$$

A density evaluates a continuous action and may exceed one; it is integrated over a region to obtain a probability. A log density sums the contributions of all $d$ latent coordinates. The index $s$ follows the inference schedule. A zero-variance step is deterministic and has no Gaussian density of this form.

# 8 DDPO and the reference reward

Denoising Diffusion Policy Optimization (DDPO) (Black et al., 2024) views a denoising trajectory as an episode. The state contains the noisy latent, timestep and prompt. The action is the next latent. A frozen scorer evaluates the decoded image at the end.

**How a final reward trains earlier steps.** Retain the previous section's latent dimension $d \in \mathbb{N}_{>0}$, conditioning $c \in \mathcal{C}$ and parameter vector $\theta \in \mathbb{R}^{d_\theta}$, with $d_\theta \in \mathbb{N}_{>0}$. An inference schedule has $J \in \mathbb{N}_{>0}$ transitions at integer indices $t_0 = T > t_1 > \cdots > t_J = 0$, where $T \in \mathbb{N}_{>0}$ is the training noise horizon. The trajectory is $\xi = (x_{t_0}, \dots, x_{t_J}) \in (\mathbb{R}^d)^{J+1}$. Let $\mathcal{S} \subseteq \{1, \dots, J\}$ index its positive-variance transitions. Their scalar log densities are $\ell_{\theta,j} = \log p_\theta(x_{t_j} \mid x_{t_{j-1}}, c) \in \mathbb{R}$. The terminal reward $R(\xi, c) \in \mathbb{R}$ and a prompt-dependent baseline $b(c) \in \mathbb{R}$ give the policy-gradient update direction

$$g(\theta) = \mathbb{E}_{\xi \sim P_\theta(\cdot|c)} \left[ (R(\xi, c) - b(c)) \sum_{j \in \mathcal{S}} \nabla_\theta \ell_{\theta,j} \right] \in \mathbb{R}^{d_\theta},$$

where $P_\theta(\cdot \mid c)$ is the probability law on trajectories in $(\mathbb{R}^d)^{J+1}$ and each $\nabla_\theta \ell_{\theta,j} \in \mathbb{R}^{d_\theta}$. Under the score-function assumptions, including a parameter-independent initial noise law and reward with no explicit parameter dependence, this is the reward gradient when all parameter-dependent sampling steps are covered. The local trainer treats saved rewards as fixed during an update and omits deterministic terminal steps. A parameter-dependent deterministic terminal map makes this an update surrogate for the full generation pipeline.

For intuition, suppose an image receives reward $0.8$ against baseline $0.5$. The positive difference $0.3$ encourages the sampled denoising actions that produced it. A below-baseline image reverses that incentive. The scorer supplies one number; learning acts through the transition log densities. Averaging many sampled trajectories supplies the statistical credit assignment.

**Reusing trajectories and measuring drift.** For a saved stochastic transition, the current-to-rollout density ratio is $u_j = \exp(\ell_{\theta,j} - \ell_{\text{old},j}) \in (0, \infty)$, with $\ell_{\text{old},j} \in \mathbb{R}$ recorded at generation. It evaluates the same saved successor under both models. DDPO uses such ratios with clipping, analogously to GRPO. The local implementation averages log densities over coordinates and therefore uses $\exp[(\ell_{\theta,j} - \ell_{\text{old},j})/d]$. This changes the scale on which clipping acts.

At one state, write $p_\theta$ and $p_{\text{ref}}$ for current and frozen-reference Gaussian transition laws on $\mathbb{R}^d$. With mean vectors $\mu_\theta, \mu_{\text{ref}} \in \mathbb{R}^d$ and equal scalar variance $\sigma^2 \in (0, \infty)$, their scalar conditional divergence is

$$\text{KL}(p_\theta \| p_{\text{ref}}) = \frac{\|\mu_\theta - \mu_{\text{ref}}\|_2^2}{2\sigma^2} \in [0, \infty).$$

The project averages this divergence over coordinates and stochastic steps to obtain $\bar{K} \in [0, \infty)$, then uses reference reward $1/(1 + \bar{K}/s_0) \in (0, 1]$ for scale $s_0 \in (0, \infty)$ fitted on training probes. Similar transition means earn a higher reward. Because a weighted reward can trade reference similarity for evasion, held-out prompt alignment is also measured directly.

# 9 Diffusion evasion: a quality-constrained result and its limits

**Question and prerequisites.** This part asks whether Stable Diffusion 1.5 (SD1.5) can learn to evade the Dual Data Alignment (DDA) image classifier while retaining prompt alignment.

**Independent detector baseline.** The official DDA checkpoint is evaluated before local adaptation. Its AUROC is 0.8221 on the ArtiFact holdout, 0.3642 on CIFAKE and 0.7716 on Shoes. Corresponding human false-positive rates are 23.71%, 87.43% and 49.82% under the recorded operating point. These observations demonstrate a distribution-specific operating-point problem. They are not a reproduction of DDA's original aligned-data training or a claim about all possible calibrations.

**Short adaptation loop.** DDA is warmed on matched initial-SD1.5 and real examples. The attacker then uses 800 fitting prompts, with 200 validation prompts and 500 evaluation prompts reserved. The bounded reward weights are 0.64 detector evasion, 0.26 CLIP alignment and 0.10 initial-model reference similarity. A training-only probe freezes the reference scale at $0.00034441052$ nats per latent coordinate; probe weights are discarded before the main run.

In the completed short run, warmed-DDA evasion is 1.0% on base outputs and 0.8% on attacked outputs. After a defender update it is 0.2% on the saved attacked outputs. This is a negative attack result under the tested recipe. It is not evidence that diffusion evasion is impossible.

**Long run and checkpoint selection.** The longer experiment generates up to 12,800 training images in rollouts of eight, with validation every 800 images. A checkpoint is eligible only if mean validation CLIP alignment remains at least 95% of the initial model's value. The selected checkpoint occurs at 5,600 training images. Selection uses 200 validation prompts; the reported final comparison uses 500 held-out test prompts and identical baseline noise seeds.

| Generator | Test evasion at .5 | CLIP | Retention |
|---|---|---|---|
| Initial SD1.5 | 1.2% | .3285 | 100% |
| Validation-selected, 5,600 images | 8.2% | .3256 | 99.1% |
| Post-hoc final, 12,800 images | 99.4% | .2080 | 63.3% |

Table 4: The post-hoc final checkpoint is an exploratory evaluation, not a replacement for the validation-selected result. CLIP retention is a ratio of mean alignment scores, not a percentage of visual quality.

The final checkpoint's high evasion at threshold 0.5 coexists with AUROC 0.9661. At the warmed defender's frozen validation threshold, approximately 0.01406, its evasion is 38.4% with 2.8% human false positives. This distinction matters: crossing a fixed threshold does not imply destroying the detector's ranking ability. Threshold choice must remain separate from test-time attack selection.

**Diagnostics and unresolved follow-up.** Saved diagnostics show zero rollout/replay log-probability discrepancy and a nonzero initial policy gradient. These are useful implementation checks, but they do not prove complete optimization correctness. In particular, the local transition log density is averaged across coordinates, which changes the scale of the importance ratio used in the update.

The follow-up long-run defender job recorded a startup failure involving an unbound receipt helper. A separate recovery directory exists, but its inspected manifest records no new defender job. The public pointer still says pending. Without a verified completion receipt, this report treats the follow-up defense outcome as unresolved rather than reporting a completed result or asserting that no other VM artifact exists.

**What was learned.** Training can push fixed-threshold evasion very high while losing prompt alignment. Enforcing a validation utility floor changes the scientific conclusion to a smaller, qualified gain. Both outcomes belong in the report because they answer different questions. The single seed and explored checkpoint trajectory do not establish a general evasion frontier or resistance to another adaptive round.

# Part III
# Text and image watermark persistence

## 10 Preservation is part of the attack objective

Removing a watermark by replacing a useful answer with nonsense, or by replacing a particular face with another face, does not solve the same problem as removing the watermark while retaining the original content. We therefore distinguish a detector outcome from a successful content-preserving attack. This distinction applies to the text and image parts independently.

Let $N \in \mathbb{N}_{>0}$ be the number of attempted inputs and $i \in \{1, \dots, N\}$ their index. The output space is $\mathcal{X} = \mathcal{V}^*$ for text over a finite vocabulary $\mathcal{V}$, or $\mathcal{X} \subseteq \mathbb{R}^{H \times W \times 3}$ for RGB images with $H, W \in \mathbb{N}_{>0}$. Original $x_i \in \mathcal{X}$ and candidate $\tilde{x}_i \in \mathcal{X}$ form a pair. A possibly randomized attacker is $A : \mathcal{X} \times \Omega \to \mathcal{X}$, with random input $\omega_i \in \Omega$, so $\tilde{x}_i = A(x_i, \omega_i)$. Model-level attacks generate these pairs from the supplied and adapted models using coupled seeds.

Let $D : \mathcal{X} \to \{0, 1\}$ be the frozen owner decision, with one meaning detected. A preservation rule $C : \mathcal{X} \times \mathcal{X} \to \{0, 1\}$ is fixed before inspecting confirmatory owner outcomes. Define scalar indicators

$$I_i = D(x_i), \qquad R_i = I_i[1 - D(\tilde{x}_i)], \qquad P_i = C(x_i, \tilde{x}_i). \tag{1}$$

Each $I_i, R_i, P_i \in \{0, 1\}$ indicates initial detection, removal from an initially detected source, or preservation, respectively. Over the $N$ attempts they form binary vectors in $\{0, 1\}^N$. The three empirical rates, each in $[0, 1]$ when its denominator is positive, are

$$\hat{r}_{\text{raw}} = \frac{\sum_{i=1}^N R_i}{\sum_{i=1}^N I_i}, \qquad \hat{r}_{\text{joint}} = \frac{\sum_{i=1}^N R_i P_i}{\sum_{i=1}^N I_i}, \qquad \hat{r}_{\text{conditional}} = \frac{\sum_{i=1}^N R_i P_i}{\sum_{i=1}^N I_i P_i}. \tag{2}$$

A zero denominator is reported as undefined. Initially undetected originals remain in the dataset and cannot count as removals. Missing or non-finite quality measurements fail the declared automatic rule.

**A concrete example.** Suppose 100 originals are detected. An attacker makes 80 undetected, but only 30 of those 80 retain the content. If 50 candidates pass preservation overall, raw removal is $80\%$, joint success is $30\%$, and conditional removal is $60\%$. Reporting only the last number would conceal that half the attempted attacks failed preservation.

**Text.** Rewriting calls for comparison with the full original answer: semantic similarity, entailment in both directions, retained numbers and entities, fluency, completion and task compliance. For Jina similarity, the two answers are encoded as unit vectors in $\mathbb{R}^{d_J}$ with encoder width $d_J \in \mathbb{N}_{>0}$; their scalar dot product lies in $[-1, 1]$. A high cosine can overlook a changed negation or a single crucial number. An automatic entailment model can also be wrong. Bidirectional natural language inference (NLI) checks whether each answer entails the other's claims. Reference perplexity is a positive scalar obtained by exponentiating mean negative token log probability under an independent model; lower values indicate text that model finds more predictable. The retained historical text diagnostic uses bidirectional NLI, a truncation check, numeric-set agreement, valid perplexity, wrapper checks and a stated Jina cosine floor. It is explicitly post hoc. It does not establish human-verified factual equivalence, nor does it supply missing original-answer perplexities retroactively.

Fresh answers to the same question are a different intervention from rewriting an existing answer. Similarity to the saved original is still reported, but answer correctness and task utility must also be assessed. A changed answer cannot be called a preserved rewrite merely because it addresses the same topic. Length caps and evaluation-tokenizer truncation are separate failure sources.

**Images.** Learned Perceptual Image Patch Similarity (LPIPS) returns a nonnegative scalar feature distance, with smaller values indicating closer images. Structural similarity (SSIM) returns a scalar in $[-1, 1]$ in its standard form, with larger values indicating closer structure. Neither has units of a percentage.

For image-image CLIP, let $f_I(x), f_I(\tilde{x}) \in \mathbb{R}^{d_e}$ be unit-normalized image embeddings of a pair $x, \tilde{x} \in \mathcal{X}$, with embedding width $d_e \in \mathbb{N}_{>0}$. Their scalar cosine is $f_I(x)^\top f_I(\tilde{x}) \in [-1, 1]$. For class $y \in \mathcal{Y}$ in the finite class set $\mathcal{Y}$, let $f_T(y) \in \mathbb{R}^{d_e}$ be the unit-normalized CLIP embedding of its class-description text. Class alignment is $a(x, y) = f_I(x)^\top f_T(y) \in [-1, 1]$. The recorded per-pair retention is the scalar $a(\tilde{x}, y)/a(x, y) \in \mathbb{R}$ when $a(x, y) > 0$. A nonpositive reference alignment makes retention undefined and fails the automatic rule. A ratio above one means the candidate aligns more strongly with the class text. Class-text CLIP checks agreement with a requested class, not retention of a particular object, identity, spatial arrangement or fine detail. A weight-modified generator may produce another valid scene from the same seed. That may preserve population quality while failing the user's stricter paired-content objective. Both conclusions should be visible rather than conflated.

**New experiments.** Preservation thresholds, candidate-selection rules and the validation/test split must be frozen before inspecting confirmatory owner scores. Threshold-sensitivity curves accompany the primary rule. Attackers may optimize quality or choose among candidates using independent quality metrics, but the final owner detector and test outcomes cannot become an unreported selection oracle. Every attempted input remains in the accounting, including rejected candidates and failed generations. Fixed-position qualitative comparisons show both successes and failures; automatic proxy passes remain subject to qualitative inspection.

**Which image thresholds apply.** The original image-development screen requires LPIPS at most $0.2$, SSIM at least $0.6$, image-image CLIP at least $0.9$, and class-alignment retention at least $0.95$, with all metrics finite. The historical joint-image diagnostic uses these same numerical bounds. Section 26 reports a later, explicitly post-hoc rule with retention at least $0.9$ and no mandatory SSIM cutoff. Each table states the protocol it uses.

# 11 Watermark testing and calibration

A keyed watermark makes generated outputs statistically depend on an owner-controlled secret while preserving their usefulness. Let $\mathcal{X}$ be the output space, either finite text sequences or RGB images, and let $k \in \mathcal{K}$ be a key in a specified key space $\mathcal{K}$. A score function $S : \mathcal{K} \times \mathcal{X} \to \mathbb{R}$ returns $S_k(x) = S(k, x) \in \mathbb{R}$ for output $x \in \mathcal{X}$. A threshold $\tau \in \mathbb{R}$ yields decision $\mathbf{1}\{S_k(x) > \tau\} \in \{0, 1\}$. The null hypothesis is that the output lacks this watermark. A false positive flags such an output; detection rate counts flagged marked outputs.

Some detectors return a scalar $p$-value in $[0, 1]$, the null tail probability of an equally or more extreme statistic. At level $\alpha \in (0, 1)$, the rule flags $p \leq \alpha$. A valid null model is needed for this rule's false-positive interpretation. AUROC measures ranking over thresholds; it does not determine detection at the chosen operating point.

For a payload of $B \in \mathbb{N}_{>0}$ bits, let $m, \widehat{m} \in \{0, 1\}^B$ be the embedded and recovered messages. Their number of matching bits is $b = \sum_{j=1}^{B} \mathbf{1}\{\widehat{m}_j = m_j\} \in \{0, \dots, B\}$, and bit accuracy is $b/B \in [0, 1]$. A payload with 70 correct bits out of 100 has 70% bit accuracy. It contributes one detected output only if it crosses the frozen decision threshold. A binomial null assumes independent balanced bit matches, an assumption that a learned extractor may violate. Calibration therefore uses separate clean examples.

With zero false positives among $n \in \mathbb{N}_{>0}$ independent null outputs, the one-sided 95% upper confidence bound on the false-positive probability is $1 - 0.05^{1/n} \in (0, 1)$. At $n = 256$ it is about $1.16\%$. A nominal $0.1\%$ target consequently needs substantially more null evidence. Calibration data and final attack tests have separate roles.

## 12 TextSeal and TextSeal 2 keyed sampling

TextSeal (Sander et al., 2026a) and TextSeal 2 (Sander et al., 2026b) alter sampling randomness to leave evidence that an owner can recompute from the resulting text. A pseudorandom function maps a secret key and token context to reproducible values. To a party without the key, those values are modeled as random.

**A race that samples the model distribution.** Let $\mathcal{V}$ be the vocabulary with $V = |\mathcal{V}| \in \mathbb{N}_{>0}$, and $h \in \mathcal{V}^*$ the current token prefix. At this fixed prefix, $p = (p_v)_{v\in\mathcal{V}} \in \Delta(\mathcal{V})$ is the next-token distribution after temperature and truncation. Each $p_v \in [0, 1]$ is a scalar. Ideal independent keyed values $U = (U_v)_{v\in\mathcal{V}} \in (0, 1)^V$ have coordinates $U_v \sim \text{Uniform}(0, 1)$. The selected token is

$$Y = \arg \min_{v\in\mathcal{V}:p_v>0} \frac{-\log U_v}{p_v} \in \mathcal{V}.$$

Think of one clock per eligible token. Its random ringing time $-\log U_v/p_v \in (0, \infty)$ has an exponential distribution with rate $p_v$. The first clock wins, and $\Pr(Y = v) = p_v$. Tokens with zero probability never win. More probable tokens have faster clocks, while the key controls their reproducible randomness.

Averaging over ideal keyed randomness preserves the token marginal. Holding the key fixed creates the dependence the detector uses. At a fixed prefix with positive probabilities, a selected token with large $U_v$ has a short clock and typically supplies stronger keyed evidence. When the null token $Y$ is independent of the keyed vector $U$, the scalar evidence $s = -\log(1 - U_Y) \in (0, \infty)$ has an exponential distribution with mean one. The sum of $n \in \mathbb{N}_{>0}$ independent such scores has Gamma distribution with shape $n$ and scale one. Repeated keyed contexts require deduplication or another dependence correction. TextSeal routes steps between two keys to restore diversity; its detector combines their evidence. Entropy weighting and searching for marked spans introduce additional calibration requirements.

**Mixing keys with fresh noise.** TextSeal 2 uses $M \in \mathbb{N}_{>0}$ key streams. For $j \in \{1, \dots, M\}$ and $v \in \mathcal{V}$, let $U_v^{(j)} \in (0, 1)$ be their ideal independent uniforms. The standard-normal cumulative distribution function is $\Phi : \mathbb{R} \to (0, 1)$, with inverse $\Phi^{-1} : (0, 1) \to \mathbb{R}$. Transform and merge the scalar values as

$$Z_v^{(j)} = \Phi^{-1}(U_v^{(j)}) \in \mathbb{R}, \qquad Z_v^\star = \frac{1}{\sqrt{M}} \sum_{j=1}^{M} Z_v^{(j)} \in \mathbb{R}.$$

The arrays $(U_v^{(j)})$ and $(Z_v^{(j)})$ have shape $M \times V$, and $Z^\star = (Z_v^\star)_{v\in\mathcal{V}} \in \mathbb{R}^V$. Independently draw fresh $\eta = (\eta_v)_{v\in\mathcal{V}} \sim \mathcal{N}(0_V, I_V)$ in $\mathbb{R}^V$, with zero vector $0_V \in \mathbb{R}^V$ and identity matrix $I_V \in \mathbb{R}^{V\times V}$. Use mixing strength $\rho \in [0, 1]$ to form

$$\widetilde{U}_v = \Phi\big(\rho Z_v^\star + \sqrt{1 - \rho^2}\eta_v\big) \in (0, 1).$$

Use $\widetilde{U}_v$ in the token race. Under the ideal independence assumptions, each mixed Gaussian has unit variance, so the transformed value is uniform when averaging over keys and noise. For a fixed key, $\rho$ controls how much keyed information survives. At $\rho = 0$, the fresh noise supplies all randomness. At $\rho = 1$, the merged key supplies it all. Intermediate values mix reproducibility with fresh variation. An owner knowing the merged key can recompute its evidence; a detector given fewer component keys has a different signal strength.

TextSeal 2 also handles speculative decoding, where a smaller draft model proposes tokens that a target model accepts or replaces. Its replacement procedure uses the memoryless property of exponential clocks. The local experiments use target-only decoding, so their results concern the sampling and detection setting described here.

# 13 Text watermarking: generation, removal and preservation

**Question and access.** A watermark differs from a content detector because its statistical signal is deliberately inserted using an owner-controlled mechanism. This part compares a released sampler watermark with a learned in-model adaptation. The reader can start here without reading the detector benchmarks or reinforcement-learning experiments.

**Two distinct watermark locations.** TextSeal 2 marks the next-token sampling process while leaving the underlying Qwen weights unchanged. The DistSeal-inspired text adaptation first trains a hidden-state embedder and text extractor, then distils teacher distributions into Qwen LoRA weights and merges the student. The latter is a local experimental adaptation of the latent teacher/student idea, not an official implementation of a published DistSeal text system. Its teacher trains for 7,500 steps, and the rank-128 student for 8,000 steps. Teacher validation bit recovery reaches 70.36%; that quantity is not the fraction of fresh student responses detected.

**Data and generation.** Both arms use Qwen3.5-4B instruct with its native chat format and thinking disabled. HC3 full questions, rather than human-answer prefixes, form prompts. Normalized question hashes separate fitting, removal, validation and test pools; duplicate questions and answer prefixes are excluded. There are 256 test questions per arm. The recorded sampling configuration uses temperature one, nucleus sampling at cumulative probability 0.95, and a 200-token cap. Nucleus sampling keeps the most likely tokens needed to reach that mass, then renormalizes their probabilities.

**The removal methods.** Ordinary rewriting and BIRA use an independent Llama-3.1-8B-Instruct rewriter with text-only access. Direct top-10 resampling and confidence smoothing instead generate fresh answers from simulated probability access. For TextSeal 2 this access reveals underlying pre-watermark logits. Sampling directly from those logits bypasses the external keyed sampler; it does not demonstrate the same bypass when a provider reveals only the keyed distribution. For the learned student, probabilities come from its merged marked weights. Smoothing combines target and independent Qwen3.5-0.8B probabilities using the adapted confidence rule.

**Detection is not calibrated identically.** TextSeal 2 uses the released v2 deduplicated Gamma detector at nominal $p \leq .001$ and, secondarily, $p \leq .01$, where $p \in [0, 1]$ is the null-tail $p$-value. The student uses extractor thresholds fitted on disjoint validation negatives. Its primary score threshold is $> -0.6263284683$. Comparing the two primary rates therefore compares two different operating-point constructions. A target-FPR label must not be presented as a precisely verified false-positive guarantee from only 256 null outputs.

**The principal result.** TextSeal 2 begins with 253/256 detections. Ordinary rewriting leaves 15 detections and BIRA leaves none. The learned student begins with only 118/256 detections, so its post-attack rates alone can overstate removal: some outputs were already negative. The portable records additionally track paired transitions among initially positive examples. Direct resampling leaves 102/256 student detections, compared with 1/256 for TextSeal 2 under the stronger underlying-logit oracle.

**Quality and semantic preservation.** Ordinary rewriting has mean Jina cosine 0.9339 and bidirectional NLI acceptance 79.7% on TextSeal 2. BIRA has cosine 0.9108 and bidirectional NLI acceptance 47.3%. A high cosine is thus compatible with an NLI model finding that the rewrite adds, drops or changes information. Numeric-set overlap is another diagnostic but does not check which number belongs to which entity or fact. Independent Mistral perplexity measures model likelihood, not truth.

A new post-hoc audit joins owner decisions, quality records and Jina scores by exact prompt ID. Its illustrative strict proxy requires bidirectional entailment, no NLI truncation, numeric-set F1 equal

| Scheme | Condition | Detected | Rate |
|---|---|---|---|
| TextSeal 2 | Original marked | 253/256 | 98.8% |
| TextSeal 2 | Clean control | 1/256 | 0.4% |
| TextSeal 2 | Ordinary rewrite | 15/256 | 5.9% |
| TextSeal 2 | BIRA | 0/256 | 0.0% |
| TextSeal 2 | Top-10 resampling | 1/256 | 0.4% |
| TextSeal 2 | Smoothing | 0/256 | 0.0% |
| Text student | Original marked | 118/256 | 46.1% |
| Text student | Clean control | 2/256 | 0.8% |
| Text student | Ordinary rewrite | 9/256 | 3.5% |
| Text student | BIRA | 1/256 | 0.4% |
| Text student | Top-10 resampling | 102/256 | 39.8% |
| Text student | Smoothing | 18/256 | 7.0% |

Table 5: Retained text detections at distinct primary operating points. Probability arms generate fresh responses; the student is a local adaptation.

to one, valid perplexity, no wrapper leakage or long letter runs, and cosine at least .90. Missing or duplicate IDs cause an error. The audit does not tune the original attacks or thresholds.

| Scheme | Rewrite | Initial + | Raw removed | Proxy pass | Joint success |
|---|---|---|---|---|---|
| TextSeal 2 | vanilla | 253 | 238 | 154 | 141 |
| TextSeal 2 | bira | 253 | 253 | 62 | 62 |
| Text student | vanilla | 118 | 110 | 152 | 60 |
| Text student | bira | 118 | 117 | 62 | 28 |

Table 6: Post-hoc joint text diagnostics, 256 outputs per arm. Proxy pass requires the stated automatic checks and Jina cosine at least .90. Joint success also requires original detection followed by non-detection. These are counts, not certified factual-preservation rates.

The strict proxy leaves 62 BIRA successes among 253 initially detected TextSeal 2 responses, versus 141 ordinary-rewrite successes. This changes the comparison substantially from raw detector suppression. These results are exploratory because the retained test outputs have already been inspected. The proxy is not a human-certified preservation criterion, and its cosine threshold is not a universal semantic constant. Confirmation experiments must freeze their criteria before producing new test outputs.

**Length limits are part of the result.** In the retained TextSeal 2 arm, 254/256 original responses reach the 200-token budget. Some source answers are therefore incomplete, and a rewrite may finish an unfinished thought rather than preserve exactly the same claims. Some NLI pairs also exceed the evaluator's 512-token context. The full records retain these cases. A future quality-controlled experiment should distinguish complete responses, truncation, factual coverage and raw detectability rather than filtering inconvenient outputs after seeing their detector scores.

## 14 DistSeal and the local text adaptation

DistSeal (Rebuffi et al., 2026) first learns to hide a bit message in an image latent, then trains a generator or decoder to reproduce the marked targets. This places learned marking behavior in model weights.

**Objects and network roles.** Let $\mathcal{X} \subseteq \mathbb{R}^{H\times W\times 3}$ be the RGB image space, with $H, W \in \mathbb{N}_{>0}$. An encoder $\mathcal{E} : \mathcal{X} \to \mathbb{R}^{d_z}$ compresses an image $x \in \mathcal{X}$ into $z = \mathcal{E}(x) \in \mathbb{R}^{d_z}$, where $d_z = C_z H_z W_z$

counts latent coordinates and $C_z, H_z, W_z \in \mathbb{N}_{>0}$ are its channel, height and width dimensions. Here the latent is continuous before any codebook quantization. Flattening changes only its representation. A decoder $\mathcal{D} : \mathbb{R}^{d_z} \to \mathcal{X}$ reconstructs an image.

For payload length $B \in \mathbb{N}_{>0}$, the message is $m \in \{0,1\}^B$. An embedder $\mathcal{W} : \mathbb{R}^{d_z} \times \{0,1\}^B \to \mathbb{R}^{d_z}$ predicts a message-dependent residual. With scalar strength $\lambda_w \in [0, \infty)$, form

$$z_w = z + \lambda_w \mathcal{W}(z, m) \in \mathbb{R}^{d_z}, \qquad x_w = \mathcal{D}(z_w) \in \mathcal{X}.$$

An extractor $\mathcal{F} : \mathcal{X} \to \mathbb{R}^B$ predicts one real logit per bit. With sigmoid $\text{sigmoid}(a) = 1/(1+e^{-a}) \in (0,1)$ for $a \in \mathbb{R}$, thresholding each logit at zero produces $\widehat{m} \in \{0,1\}^B$. For example, payload length $B = 3$ with message $(1,0,1) \in \{0,1\}^3$ asks the embedder to alter the latent so that the extractor later predicts those bits. It adds no visible text label.

Post-hoc training combines recovery after image augmentations with a pixel-space adversarial discriminator. Quantization converts a continuous latent to discrete codebook tokens. Embedding before or after that conversion changes which targets are available for generator training. A straight-through gradient supplies a training surrogate through the discrete step.

**What distillation transfers.** Generator distillation fits marked latents with the architecture's native objective, such as categorical prediction for RAR or denoising for DC-AE. Decoder distillation trains a decoder receiving the original latent to reconstruct the teacher-marked image. For a fixed release payload, the student can then produce marked outputs directly. Later clean-data fine-tuning can change this learned behavior, which motivates the removal experiments.

**The local text version.** The text prototype applies an embedder to hidden states, then distils teacher token probabilities into Qwen LoRA weights. Let $\mathcal{V}$ be its vocabulary of size $V \in \mathbb{N}_{>0}$, $c \in \mathcal{V}^*$ a prompt, and $y = (y_1, \ldots, y_L) \in \mathcal{V}^L$ a training response of length $L \in \mathbb{N}_{>0}$. Write its hidden-state matrix as $H_{\text{resp}} \in \mathbb{R}^{L \times d_h}$, with hidden width $d_h \in \mathbb{N}_{>0}$. The corresponding teacher and student token-logit matrices are $\Lambda_T, \Lambda_\theta \in \mathbb{R}^{L \times V}$. The subscript $T$ denotes the teacher. A text extractor $\mathcal{F}_{\text{text}} : \mathcal{V}^* \to \mathbb{R}^B$ maps a finite token sequence to its message logits.

At response position $t \in \{1, \ldots, L\}$, let $p_T(\cdot \mid c, y_{<t})$ and $p_\theta(\cdot \mid c, y_{<t})$ be teacher and student probability vectors in $\Delta(\mathcal{V})$, with student parameters $\theta \in \mathbb{R}^{d_\theta}$ for $d_\theta \in \mathbb{N}_{>0}$. Each probability vector is the vocabulary softmax of its corresponding real-valued logit row, so its theoretical coordinates are strictly positive. A summed forward-KL distillation loss is

$$L_{\text{student}}(\theta) = \sum_{t=1}^{L} \sum_{v \in \mathcal{V}} p_T(v \mid c, y_{<t}) \log \frac{p_T(v \mid c, y_{<t})}{p_\theta(v \mid c, y_{<t})} \in [0, \infty).$$

The loss penalizes the student for assigning too little probability to tokens favored by the teacher. Teacher forcing supplies the saved training tokens as the prefix at each position, so both models see the same prefix. Response masks exclude prompt and padding positions, and the training reduction may average eligible positions. In free generation the student visits its own prefixes, so teacher recovery alone cannot establish detection on fresh answers. This is a local text adaptation of the latent teacher/student idea.

# 15 Image watermarking: weight changes versus output purification

**Question and prerequisites.** This part asks whether a watermark survives changes to a released generator or transformations of one generated image. It can be read independently of text watermarking. The key distinction is that a model-level intervention changes future generations, while an output-level intervention changes only the supplied image.

| Model | Condition | Detected | LPIPS | Class CLIP |
|---|---|---|---|---|
| RAR | Marked original | 100.0% | 0.0000 | 0.3030 |
| RAR | Clean LoRA 2,500 | 4.9% | 0.6692 | 0.3044 |
| RAR | Clean full 2,500 | 7.2% | 0.6618 | 0.3032 |
| RAR | NFPA 40 | 21.5% | 0.4221 | 0.2630 |
| DCAE | Marked original | 100.0% | 0.0000 | 0.2934 |
| DCAE | Clean LoRA 2,500 | 0.2% | 0.1510 | 0.2840 |
| DCAE | Clean full 2,500 | 0.4% | 0.2148 | 0.2816 |
| DCAE | NFPA 40 | 33.6% | 0.5082 | 0.2664 |

Table 7: Matched image measurements, 512 outputs per condition. LPIPS measures paired drift; class CLIP is an incomplete utility proxy.

**Native models and independent fitting.** The experiments use released DistSeal RAR-XL at 256 pixels and UViT-H/DC-AE at 512 pixels, retaining their original decoders and owner extractors. RAR predicts discrete image tokens autoregressively. UViT-H is a diffusion generator over continuous DC-AE latents; DC-AE supplies the image encoder and decoder. Each primary condition has 512 marked test outputs, 512 clean-generator controls and 256 clean calibration outputs. Current clean-data attacks fit on 8,000 ImageNet training images from 800 classes, with 800 separate validation images. The remaining 200 classes are excluded from attack fitting, and the original synset ordering is preserved.

The historical pilot instead uses 50 images of a single class. Its outcomes differ sharply from the broad-fitting experiment, especially for DC-AE. They remain separate protocols in the detailed tables. A similar method name does not justify pooling their measurements.

**Weight access.** Merging uses the compatible original clean checkpoint and predetermined marked fractions 0.75, 0.5, 0.25 and zero. Full clean replacement is a sanity control requiring stronger access than possession of marked weights alone. Continued clean LoRA or full training uses the generator's native objective without owner scores or clean reference weights. Task schemas, frozen artifacts and separate owner evaluation constrain this research workflow's inputs; they are not an operating-system security boundary against arbitrary code on the same account.

**Detection and sample drift.** After 2,500 clean LoRA updates, detection falls from 100% to 4.9% for RAR and 0.2% for DC-AE. Full updates yield 7.2% and 0.4%. RAR's class-text CLIP remains close to its baseline mean even though LPIPS between paired seed outputs is large. Changing autoregressive weights can change which valid scene is sampled. Thus paired LPIPS measures sample drift and cannot by itself establish poor population-level generation quality.

DC-AE partial merging provides a different warning. At 75% marked weights, detection falls to zero, but LPIPS is 0.8684 and class-text CLIP falls from 0.2934 to 0.2141. That is suppression with a substantial measured utility cost. Clean LoRA is much closer to the original samples under the recorded perceptual measurements.

**Output-only NFPA.** NFPA receives saved image outputs and an independent public SD2.1 model. It receives neither target weights nor owner detector feedback. The experiment simulates an output-only provider; it does not query a commercial service. Strengths zero, 20 and 40 are fixed in advance, with 40 the primary setting. Zero strength still performs regeneration; a separate resize-only arm measures the resizing intervention.

At strength 40, 110/512 RAR images and 172/512 DC-AE images remain detected: 21.5% and 33.6%. LPIPS is 0.4221 and 0.5082, respectively. Class-text CLIP falls to 0.2630 and 0.2664, compared with original means 0.3030 and 0.2934. Qualitative examples show changes to faces, objects and framing. This is neither universal removal nor exact content preservation.

**Matched quality, not incomparable score names.** The later quality extension verifies that all source reference PNGs are bit-identical across the weight and NFPA experiments, then measures the same metrics for all 13 conditions per model: 13,312 paired evaluations. It reproduces earlier overlapping metrics within a maximum absolute discrepancy of $2.3842 \times 10^{-7}$. No attack or model training is repeated for that extension.

Image-image CLIP and class-text CLIP answer different questions. SSIM and PSNR depend strongly on alignment; LPIPS is a perceptual distance, not a percentage. None provides an automatic guarantee that every identity or detail survives. New attack comparisons therefore need joint preservation-qualified outcomes, with thresholds fixed independently of the final owner scores and with failures retained.

**Simple transforms and residual scope.** JPEG quality 50 retains 100% detection on both unmodified marked models in the recorded experiment. A center crop retaining half the area reduces detection to 0.8% for RAR and 1.0% for DC-AE under separately calibrated transformed-null thresholds. These transform results do not include matched perceptual-quality measurements and must not inherit them from the untransformed images. They motivate robust synchronization or augmentation research, but do not establish that cropping preserves every requested object.

## 16 Joint removal and preservation on retained image pairs

The following diagnostic reuses the complete paired quality measurements and frozen owner decisions. No image generation or attack training is repeated. Every score is joined by its original sample index; output-only cases are additionally matched through the original PNG hash and the sealed NFPA input identifier. The reconstructed owner decisions must reproduce the recorded detection counts before they enter this table.

| Model | Condition | Initial + | Raw removed | Proxy pass | Joint |
|---|---|---|---|---|---|
| RAR | weight_unmodified_supplied | 512 | 0 | 512 | 0 |
| RAR | weight_merge_0.75 | 512 | 16 | 0 | 0 |
| RAR | weight_merge_0.5 | 512 | 262 | 0 | 0 |
| RAR | weight_merge_0.25 | 512 | 498 | 0 | 0 |
| RAR | weight_merge_0 | 512 | 508 | 0 | 0 |
| RAR | weight_clean_lora_500 | 512 | 443 | 0 | 0 |
| RAR | weight_clean_lora_2500 | 512 | 487 | 0 | 0 |
| RAR | weight_clean_full_500 | 512 | 326 | 0 | 0 |
| RAR | weight_clean_full_2500 | 512 | 475 | 0 | 0 |
| RAR | output_resize_only | 512 | 0 | 511 | 0 |
| RAR | output_nfpa_0 | 512 | 22 | 105 | 3 |
| RAR | output_nfpa_20 | 512 | 157 | 2 | 1 |
| RAR | output_nfpa_40 | 512 | 402 | 0 | 0 |
| DCAE | weight_unmodified_supplied | 512 | 0 | 512 | 0 |
| DCAE | weight_merge_0.75 | 512 | 512 | 0 | 0 |
| DCAE | weight_merge_0.5 | 512 | 512 | 0 | 0 |
| DCAE | weight_merge_0.25 | 512 | 512 | 0 | 0 |
| DCAE | weight_merge_0 | 512 | 512 | 172 | 172 |
| DCAE | weight_clean_lora_500 | 512 | 504 | 298 | 293 |
| DCAE | weight_clean_lora_2500 | 512 | 511 | 284 | 283 |
| DCAE | weight_clean_full_500 | 512 | 487 | 163 | 155 |
| DCAE | weight_clean_full_2500 | 512 | 510 | 150 | 149 |
| DCAE | output_resize_only | 512 | 0 | 512 | 0 |
| DCAE | output_nfpa_0 | 512 | 76 | 83 | 12 |
| DCAE | output_nfpa_20 | 512 | 162 | 1 | 1 |

| Model | Condition | Initial + | Raw removed | Proxy pass | Joint |
|---|---|---|---|---|---|
| DCAE | output_nfpa_40 | 512 | 340 | 0 | 0 |

Post-hoc image proxy: LPIPS $\leq .2$, SSIM $\geq .6$, image-image CLIP $\geq .9$ and class-text CLIP retention $\geq .95$. All 512 original outputs per model remain in the denominator; joint success also requires initial detection and final non-detection. Failure counts and LPIPS-floor sensitivity appear in the machine-readable receipt.

The strict rule exposes a distinction hidden by raw detection rates. RAR clean-LoRA candidates frequently remain aligned with the requested class but change the individual scene, so none passes this paired-content rule. DC-AE clean LoRA preserves many original images while removing the signal: at 2,500 steps, 283 of 512 attempted inputs are joint successes, compared with 511 raw removals. At NFPA strength 40, neither model has a joint success under this rule. Lower NFPA strength retains some candidates; regeneration is therefore neither uniformly unsuccessful nor uniformly content preserving.

These thresholds are a post-hoc diagnostic on already inspected results. They do not provide an untouched benchmark, a human-quality guarantee, or evidence that RAR is robust to every quality-preserving attack. The receipt retains alternative LPIPS limits and per-metric failure counts. Subsequent confirmatory experiments must specify their primary rule on independent validation data and retain continuous metrics and qualitative failures alongside these counts.

# Part IV
# Watermark robustness experiments

## 17 Filtering the watermark update

A marked generator and its compatible clean generator have parameter vectors in $\mathbb{R}^{d_\theta}$, with $d_\theta \in \mathbb{N}_{>0}$. Let $\delta_0 \in \mathbb{R}^{d_\theta}$ be marked minus clean parameters. To understand how clean training may erase this difference, consider the quadratic loss

$$L_c(\theta) = \tfrac{1}{2}(\theta - \theta_\star)^\top H(\theta - \theta_\star) \in \mathbb{R}, \qquad \theta, \theta_\star \in \mathbb{R}^{d_\theta},$$

where $\theta_\star$ is a minimizer and the symmetric positive semidefinite Hessian is $H \in \mathbb{R}^{d_\theta \times d_\theta}$. If both generators take the same gradient steps with scalar learning rate $\eta \in (0, \infty)$, their difference after $K \in \mathbb{N}_0$ steps is

$$\delta_K = (I_{d_\theta} - \eta H)^K \delta_0 \in \mathbb{R}^{d_\theta},$$

with identity matrix $I_{d_\theta} \in \mathbb{R}^{d_\theta \times d_\theta}$.

An eigenvector direction of $H$ with scalar eigenvalue $h \in [0, \infty)$ retains the factor $(1 - \eta h)^K$. For $0 \leq \eta h \leq 1$, larger curvature means faster decay. This motivates placing watermark changes in less sensitive directions. The DC-AE pilot approximates sensitivity using clean-loss gradient energy per parameter block, then filters the marked-minus-clean update. That proxy and a nonlinear generator can depart substantially from the quadratic model. Initial detection and paired-content checks test whether the approximation produces a usable candidate.

## 18 A failed post-distillation survival-filter pilot

This new development experiment asks whether a low-cost parameter-space intervention can retain an existing watermark while redistributing it toward directions predicted to resist clean training. Section 17 explains the approximation. This is a negative result: the candidates fail before any removal attack is attempted.

The source model is the released DistSeal UViT-H/DC-AE generator, with its fixed clean decoder and owner extractor. Thirty-two clean fitting images supply native diffusion-loss gradients. Their per-parameter-tensor squared-gradient means provide sensitivity proxies. The experiment compares the original marked checkpoint, inverse-sensitivity filtering with horizon zero, and two positive-horizon filters. All filtered task vectors have the same Euclidean energy as the original marked-minus-clean update. Equal parameter energy is a control, not a guarantee of equal utility.

Evaluation uses 32 classes from the original 200-class fitting holdout, deterministically selected in hash order, and fresh coupled generation seeds. No fitting class enters this pilot evaluation. The protocol freezes a development screen requiring at least 28/32 initial detections and 28/32 paired-preservation passes before any removal attack. The preservation rule uses LPIPS, SSIM, image-image CLIP and class-text CLIP retention with the limits stated in Section 10. No candidate is selected using a confirmatory test set.

| Candidate | Detected | Preservation pass | Mean bit acc. | LPIPS |
|---|---|---|---|---|
| original | 32/32 | 32/32 | 0.9966 | 0.0000 |
| inverse | 0/32 | 0/32 | 0.5112 | 0.9295 |
| survival1 | 0/32 | 0/32 | 0.4995 | 0.9657 |
| survival4 | 0/32 | 0/32 | 0.5098 | 0.9647 |

Table 9: New DC-AE development pilot, before any removal attack. All filtered candidates fail the initial detection and paired-preservation screen. No robustness comparison is performed on them.

The original model passes both screens on every evaluated output. All three filtered models lose the watermark and fail paired preservation on every output. Their mean bit accuracies lie near chance. Consequently, clean-update and purification attacks are not run on these candidates. This rejects this blockwise, post-hoc approximation at these settings; it does not disprove the local quadratic identity or every use of curvature during training. The mathematical surrogate does not capture the model's nonlinear generation path well enough to justify this intervention.

The allocation completed in 205 seconds on one H100, including startup and quality evaluation. This development result is retained in the report and compute ledger. It is not evidence of a successful new robust watermark.

# 19 Distilling from a reference after clean adaptation

The adapted-reference experiment asks whether the student can change its output distribution under clean training while retaining its watermark. Let $\mathcal{C}$ be the conditioning space, and let $G_\theta : \mathbb{R}^{d_u} \times \mathcal{C} \to \mathbb{R}^{d_z}$ generate a flattened latent from random input $u \in \mathbb{R}^{d_u}$ and condition $c \in \mathcal{C}$. The dimensions $d_u, d_z \in \mathbb{N}_{>0}$ count random-input and latent coordinates. The parameter vector is $\theta \in \mathbb{R}^{d_\theta}$, with $d_\theta \in \mathbb{N}_{>0}$, and $\theta_0 \in \mathbb{R}^{d_\theta}$ denotes the clean reference.

A fixed support dataset is $\mathcal{B} = ((x_i, c_i))_{i=1}^{N_s} \in (\mathcal{X} \times \mathcal{C})^{N_s}$, where $N_s \in \mathbb{N}_{>0}$ and $\mathcal{X} \subseteq \mathbb{R}^{H \times W \times 3}$ is the RGB training-image space with $H, W \in \mathbb{N}_{>0}$. With optimizer, update count and random seeds fixed, the clean adaptation map $\mathcal{A}_\mathcal{B} : \mathbb{R}^{d_\theta} \to \mathbb{R}^{d_\theta}$ returns adapted parameters. For owner key $k \in \mathcal{K}$ in the scheme's key space $\mathcal{K}$, let $W_k : \mathbb{R}^{d_z} \to \mathbb{R}^{d_z}$ be the latent watermarking map with its payload and strength fixed. The training target is

$$G_{\mathcal{A}_\mathcal{B}(\theta)} \approx W_k \circ G_{\mathcal{A}_\mathcal{B}(\theta_0)}.$$

Both sides map random inputs and conditions to marked latent outputs; $\circ$ denotes function composition. The approximate equality is the desired matching behavior, assessed through native training losses and output evaluation.

For example, if clean adaptation changes a visual style, the adapted teacher first learns that style and then marks its generated latents. The fixed-reference control instead uses $W_k \circ G_{\theta_0}$ as its target after

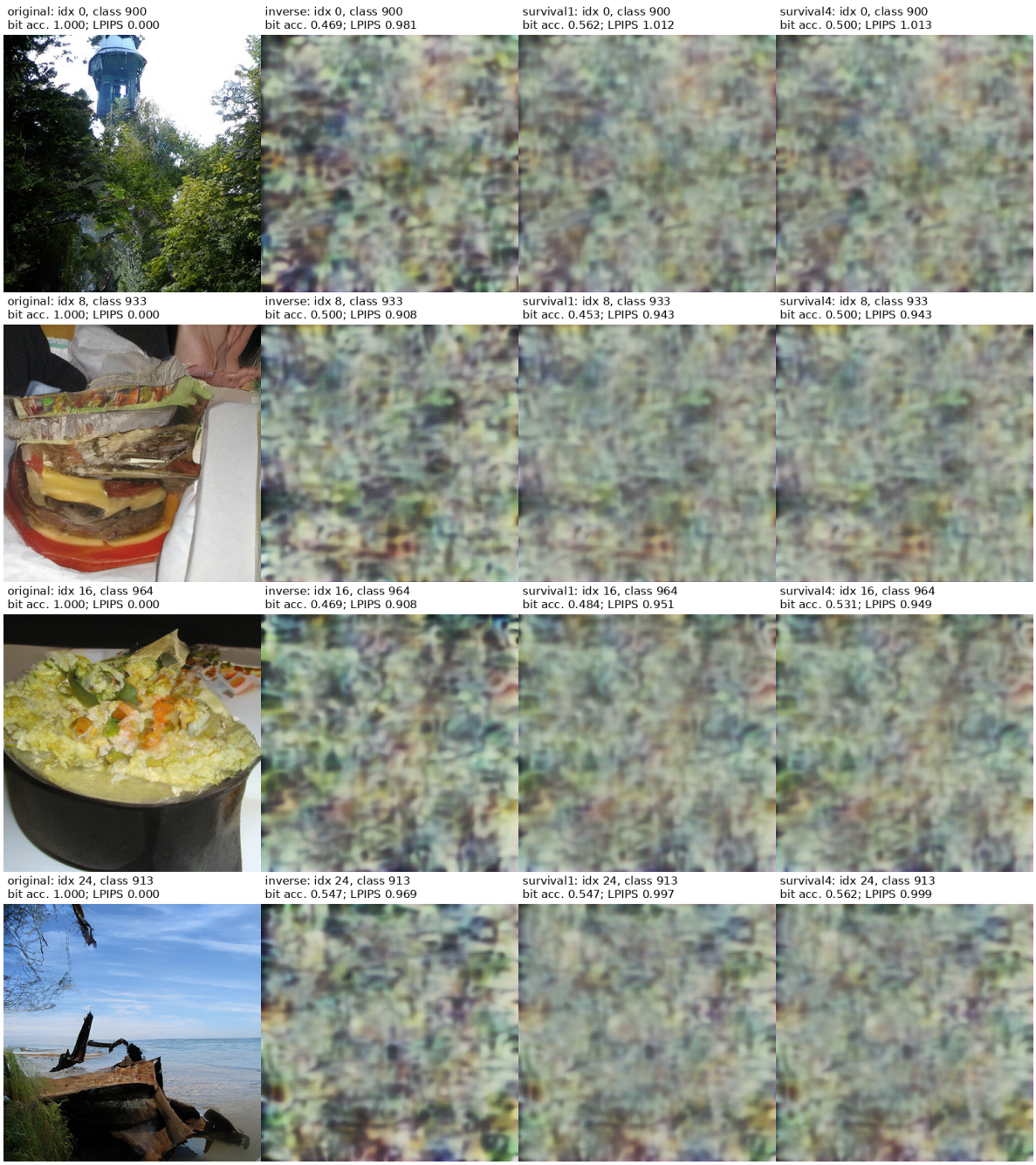



Figure 1: Four coupled pilot examples, one row per original index and one column per candidate. Positions 0, 8, 16 and 24 are displayed without filtering by outcome; this display rule was chosen after the run. Native PNGs remain unchanged. The images expose content failures that detector suppression alone would conceal.

the same student adaptation. The comparison changes which marked target the adapted student learns from. Both variants use native denoising losses on their teacher latents and a retention term on the pre-adaptation student. A first-order outer update treats the inner optimizer's derivative as the identity. Initial teacher quality and independent removal resistance are evaluated in successive experiments.

## 20 Checking an adapted-reference watermark teacher

The failed parameter-filter pilot motivates a function-space alternative: train the marked student after adaptation toward a marked version of an adapted clean reference. Section 19 defines the adapted-reference target and its fixed-reference control. Before robustness training, a prerequisite is to check whether this teacher itself produces detectable, content-preserving marks.

We use the released clean UViT-H generator, DC-AE decoder and DistSeal latent embedder. The public model card supplies watermark strength $0.5$, applied in raw autoencoder latent units before decoding. Diffusion-generator latents are divided by $0.2889$ before embedding. The fixed teacher marks clean-model samples. The adapted teacher marks samples after 100 clean AdamW updates at learning rate $10^{-5}$, batch size four, using a frozen sequence drawn from the first 2,000 of the existing 8,000 fitting images. The owner detector does not enter this clean update. Indices 4,000–7,999 are reserved for later development removal training; teacher support does not overlap them. These partitions are new to this development comparison, not a claim of an untouched external benchmark.

Thirty-two fresh coupled generation seeds and labels from the existing 200-class fitting holdout are frozen before generation. The paired-quality reference is each teacher's own clean output, before applying the embedder. The initial screen requires at least 28/32 detections and 28/32 preservation passes per teacher, under the LPIPS, SSIM and CLIP limits in Section 10. The original owner threshold $> 0.6875$ remains fixed.

| Reference | Clean detected | Marked detected | Preserved | Mean LPIPS |
|---|---|---|---|---|
| Fixed | 0/32 | 32/32 | 31/32 | 0.0411 |
| Adapted | 0/32 | 32/32 | 31/32 | 0.0396 |

Table 10: Prerequisite teacher validation. Preservation compares the post-hoc marked image with its corresponding clean reference. These counts do not measure robustness of a distilled generator after a removal attack.

Both teachers pass the screen. All 128 native PNG hashes, sample indices, class labels and preservation sums are verified against the saved records. The allocation uses 622 seconds on one H100, including a slow dependency startup and quality evaluation. A matched first-order meta-distillation comparison is launched separately; a trained model's initial quality, independent removal attacks, calibration and repeated-seed confirmation still require evaluation. No robustness gain is inferred from the table above.

## 21 Initial meta-distillation: detection survives, paired quality does not

After the teacher check, a matched development run compares fixed and adapted marked targets. Each student starts from the released marked UViT-H checkpoint. One outer iteration runs 100 detached clean AdamW updates with learning rate $10^{-5}$ and batch size four, then evaluates native denoising loss on the designated marked-teacher latent data. The clean support sequence and noise seeds exactly match those used to adapt the clean reference. The first-order meta-gradient ignores differentiation through Adam. After restoring the pre-adaptation parameters, an additional native denoising retention loss on fixed marked-teacher samples contributes with weight one. Outer AdamW uses learning rate

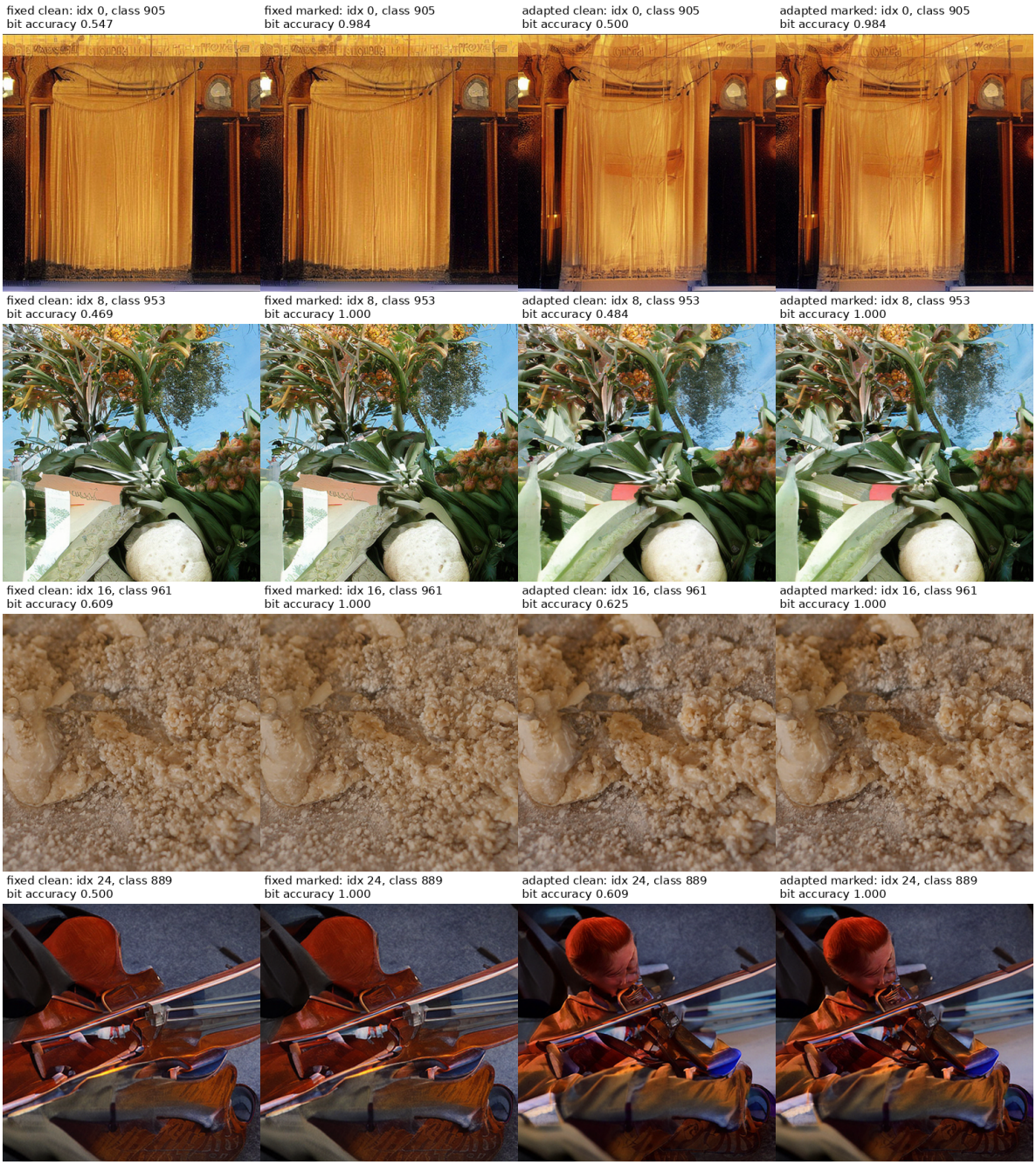



Figure 2: The fixed and adapted references before and after post-hoc marking. Rows show positions 0, 8, 16 and 24, chosen after the run without filtering by detection or quality. The display is resized uniformly; native outputs are retained. Similarity of marking pairs is a different question from similarity between the two independently adapted generators.

$2 \times 10^{-6}$ for 100 iterations. Each teacher dataset contains 256 generated latent examples with fitting-class labels. The training comparison changes only the fixed versus adapted query target; it does not claim generalization across support tasks from a single frozen adaptation sequence.

Both runs complete. The fixed-target allocation takes 825 seconds and the adapted-target allocation 703 seconds, each on one H100. Timing includes startup and target generation and is not an algorithmic speed comparison. A separate initial screen uses 32 fresh coupled seeds and fitting-holdout classes, comparing both trained models with the unchanged released marked generator. The owner threshold and all paired-preservation limits remain frozen.

| Checkpoint | Detected | Preserved | Mean bit accuracy | Mean LPIPS |
|---|---|---|---|---|
| Unchanged | 32/32 | 32/32 | 0.9937 | 0.0000 |
| Fixed target | 32/32 | 24/32 | 0.9961 | 0.1219 |
| Adapted target | 32/32 | 20/32 | 0.9971 | 0.1254 |

Table 11: Initial generation screen after 100 outer updates. Both trained models fail the requirement of at least 28/32 preservation passes, despite retaining watermark detection. No removal attack is run on them.

This failure illustrates why mean bit accuracy alone is inadequate. The new models still emit detectable marks, but do not meet the paired-content constraint required for the next stage. All 96 native PNG hashes, sample indices, labels and preservation sums are verified. The result is retained as a development failure; neither a robust defense nor a benefit of the adapted target is established.

**A revised retention target.** Denoising retention on finite synthetic data can exert a nonzero gradient even at the original model. The revision instead matches the student's predictions to those of a frozen marked checkpoint. Let $\theta, \theta_0 \in \mathbb{R}^{d_\theta}$ be the student and original marked parameter vectors, with $d_\theta \in \mathbb{N}_{>0}$. For a noisy latent $z_t \in \mathbb{R}^{d_z}$, noise level $t \in \{1, \dots, T\}$ and class condition $y \in \mathcal{Y}$, both denoisers return vectors $\varepsilon_\theta(z_t, t, y), \varepsilon_{\theta_0}(z_t, t, y) \in \mathbb{R}^{d_z}$. Here $d_z, T \in \mathbb{N}_{>0}$ count latent coordinates and training noise levels. The finite conditioning set $\mathcal{Y}$ includes the null class when class dropout is used.

The scalar prediction-retention loss is

$$\mathcal{R}_{\text{pred}}(\theta) = \mathbb{E}_{(z_t,t,y)\sim P_{\text{train}}}\left[\frac{1}{d_z}\|\varepsilon_\theta(z_t,t,y) - \varepsilon_{\theta_0}(z_t,t,y)\|_2^2\right] \in [0,\infty), \tag{3}$$

Here $P_{\text{train}}$ is the prescribed probability law on $\mathbb{R}^{d_z} \times \{1, \dots, T\} \times \mathcal{Y}$, obtained from teacher latents, sampled noise levels, Gaussian noise and class conditions. The code averages squared errors over batch samples and latent coordinates, matching the $1/d_z$ reduction. Identical predictions give zero loss and zero gradient.

For local perturbation $\delta = \theta - \theta_0 \in \mathbb{R}^{d_\theta}$, let $J(z_t, t, y) = \partial\varepsilon_\theta(z_t, t, y)/\partial\theta|_{\theta=\theta_0} \in \mathbb{R}^{d_z \times d_\theta}$ be the prediction Jacobian. A first-order expansion of the predictions gives the quadratic approximation

$$\mathcal{R}_{\text{pred}}(\theta_0 + \delta) \approx \frac{1}{d_z}\mathbb{E}\|J\delta\|_2^2 = \frac{1}{d_z}\delta^\top \mathbb{E}[J^\top J]\delta.$$

The expectation is over the same $P_{\text{train}}$, $J\delta \in \mathbb{R}^{d_z}$ and $\mathbb{E}[J^\top J] \in \mathbb{R}^{d_\theta \times d_\theta}$. Thus the anchor penalizes parameter changes according to their effect on denoiser predictions at training inputs. It supplies a local prediction constraint; paired output quality is still evaluated independently. The revision also raises its scalar coefficient to 32, so comparison with the failed run combines a target change and a coefficient change. Both query-target variants share this revision.

The revised runs complete in 797 and 733 allocation seconds for fixed and adapted targets, respectively. The fresh screen verifies all 96 native PNG hashes, labels and indices. Passing permits the next experiment; it does not demonstrate robustness. A blind attacker subsequently trains on disjoint

| Checkpoint | Detected | Preserved | Mean bit accuracy | Mean LPIPS |
|---|---|---|---|---|
| Unchanged | 32/32 | 32/32 | 0.9956 | 0.0000 |
| Fixed target | 32/32 | 32/32 | 0.9951 | 0.0077 |
| Adapted target | 32/32 | 31/32 | 0.9956 | 0.0075 |

Table 12: Revised prediction-anchor screen, with a newly frozen seed set. Both trained models pass the initial detection and preservation requirements. These are initial outputs, before an independent removal attack.

clean fitting images, receiving neither the owner detector nor the original clean model. Its denoising objective is augmented by a prediction anchor to a frozen copy of the supplied marked checkpoint, with coefficients $0, 1, 4, 16$. All four 500-update runs and their 25-, 100- and 500-update exports are retained for each of the three starting models. Quality-only validation selects an export before owner evaluation on separately reserved classes and seeds. Sections 22–26 report the completed removal evaluations.

## 22 Independent image attack: the initial comparison is inconclusive

After both prediction-anchored models pass the initial quality screen, independent attackers run 500 clean AdamW updates on 4,000 fitting images from classes 400–799. The simulated adaptation used 2,000 different images from classes 0–199. Both image identities and class labels are disjoint between those support and attack pools. The reserved final classes are outside all 800 fitting classes; they are not claimed to be unseen during ImageNet pretraining or earlier project experiments.

The attacker receives the supplied marked model and a frozen copy of it as a prediction anchor, but neither the owner's detector and message nor the original clean weights. It trains with anchor coefficients $0, 1, 4, 16$ and saves updates 25, 100 and 500. All 12 candidates per model are evaluated on 32 coupled validation seeds using only LPIPS, SSIM, image CLIP similarity and class-alignment retention. The frozen rule selects the largest update count with at least 28/32 quality passes, then the smallest anchor coefficient. It chooses update 25 with coefficient 16 for the released model, and update 25 with coefficient 4 for each trained model. The selections are frozen before generation and detector scoring on 64 different final seeds.

| Starting model | Updates | Anchor | Preserved | Raw removed | Joint removed |
|---|---|---|---|---|---|
| Original | 25 | 16.0 | 60/64 | 0/64 | 0/64 |
| Fixed | 25 | 4.0 | 56/64 | 0/64 | 0/64 |
| Adapted | 25 | 4.0 | 57/64 | 0/64 | 0/64 |

Table 13: Frozen quality-only-selected image attacks on 64 reserved final seeds and fitting-holdout classes. Each attack is paired with its own supplied marked checkpoint. The distinct selected anchor weights reflect the prespecified selection policy; they are not matched-coefficient comparisons.

All three initial checkpoints are detected on all 64 final outputs. None of the selected attacks removes a watermark, even before enforcing quality. Final preservation is 60/64 for the released model, 56/64 for the fixed-target model and 57/64 for the adapted-target model. The clean-model diagnostic is 0/64 detected. This does not establish a defense benefit: the selected attacker is ineffective against the original model too. The finite null set also cannot precisely certify a $1\%$ false-positive rate. All 384 original/attacked native PNG hashes, sample indices, labels, selection hashes and final quality hashes are checked in the analysis receipt.

**A stronger client with explicit quality fallback.** A subsequent development protocol allows an attacker to retain its supplied marked source model at inference. It generates a source and an adapted output with coupled randomness, computes the frozen quality metrics without owner feedback, and returns the source whenever the adaptation fails quality. This hybrid client pays for both generation paths and quality evaluation. It is distinct from releasing a single adapted checkpoint. A 500-update adaptation can therefore be tested without treating its damaged samples as successful removals. Validation and final generation use fresh seeds; class identities and existing training exports are reused. This revision is frozen after the negative result above, and requires its own reporting and later confirmation.

## 23 A stronger image client reveals no defense gain

The first quality-selected image attack in Section 22 cannot distinguish the models because it removes no watermark from the released model. A second development client retains its supplied marked source generator at inference, generates a source and an adapted output with coupled randomness, and evaluates their quality without owner feedback. If the adaptation fails any frozen preservation gate, the client returns the source. This costs two generation paths and quality evaluation, and is distinct from releasing one adapted checkpoint.

All three quality-only validation selections choose 500 clean updates with zero prediction anchor. Before fallback, validation quality passes are 13/32 for the released model, 14/32 for the fixed-target model and 13/32 for the adapted-target model. Fallback makes every returned validation sample preserve its source. Final generation uses 64 fresh seeds; class identities are reused, so this is seed replication rather than a new class-disjoint study.

| Starting model | Updates | Anchor | Preserved | Raw removed | Joint removed |
|---|---|---|---|---|---|
| Original | 500 | 0.0 | 64/64 | 20/64 | 20/64 |
| Fixed | 500 | 0.0 | 64/64 | 22/64 | 22/64 |
| Adapted | 500 | 0.0 | 64/64 | 22/64 | 22/64 |

Table 14: Frozen quality-only-selected image attacks on 64 reserved final seeds and fitting-holdout classes. Each attack is paired with its own supplied marked checkpoint. The distinct selected anchor weights reflect the prespecified selection policy; they are not matched-coefficient comparisons. The hybrid client returns the marked source on a quality failure; these are gated outputs, not a single adapted checkpoint. Class identities are reused with fresh generation seeds.

Before fallback, raw removal counts are 54/64, 52/64 and 55/64, while preservation counts are 24/64, 27/64 and 26/64. Their intersections are 20/64, 22/64 and 22/64. Returning the detected source on a quality failure reduces removals to those intersections and preserves all 64 returned outputs. Neither trained variant improves on the released model. These are development outcomes; they provide no positive robustness evidence for the training method.

This failure motivates one prespecified horizon revision: increase both teacher adaptation and the student's simulated inner attack from 100 to 500 updates, keeping the 100 outer updates and prediction-retention coefficient 32. The new teacher passes its prerequisite screen: 32/32 marked detections for both targets, with 30/32 fixed-target and 31/32 adapted-target preservation passes. That prerequisite establishes target quality, not robustness of a trained student. If the revised students again show no benefit under the independent attack, this meta-distillation direction stops rather than expanding to more watermark families.

## 24 What the image fallback rejects

The final client in Section 23 always returns an output that passes its frozen paired-content gates, because it can retain the supplied source generator and fall back to its output. This does not mean the raw adapted generator always preserves the source. The figure illustrates that distinction: left is the supplied source output, middle is the raw 500-update adaptation, and right is the returned quality-gated output. Source positions 0, 16, 32 and 48 are fixed by position after the run, without outcome filtering. All source, raw and returned PNG hashes and raw quality hashes are checked against the final artifact freeze. A failed raw adaptation remains visible even when the client returns the unchanged source.

Raw preservation requires all four gates: LPIPS at most $0.2$, SSIM at least $0.6$, image-image CLIP cosine at least $0.9$, and class-alignment retention at least $0.95$. A low LPIPS alone therefore does not imply a pass. The bit-accuracy labels are rounded for display; owner decisions use their full values and the frozen strict threshold $> 0.6875$.

The fixed-target and adapted-target galleries use the same positions and are in `tex/figures/`.

## 25 Matching the image adaptation horizon does not recover robustness

The final prespecified revision increases both clean-teacher adaptation and the student's simulated adaptation from 100 to 500 inner updates. The fixed-reference and adapted-reference models retain the same 100 outer updates and prediction-retention coefficient 32. All three supplied models pass the fresh initial screen, with 32/32 detections and preservation passes. This allows a matched attack comparison; it does not itself establish robustness.

The blind attacker uses clean data outside the defense support, with checkpoints at 25, 100 and 500 updates and prediction-anchor weights 0, 1, 4 and 16. Quality-only selection on 32 validation images chooses the 500-update, zero-anchor checkpoint for each supplied model. Independent owner evaluation then uses 64 final generation seeds and fitting-holdout classes, with the previously frozen detector and preservation thresholds. Validation and final class sets are disjoint; their identities were used in earlier development studies, so this is not new-class confirmation. No owner scores enter checkpoint selection.

| Starting model | Updates | Anchor | Preserved | Raw removed | Joint removed |
|---|---|---|---|---|---|
| Original | 500 | 0.0 | 64/64 | 20/64 | 20/64 |
| Fixed | 500 | 0.0 | 64/64 | 21/64 | 21/64 |
| Adapted | 500 | 0.0 | 64/64 | 20/64 | 20/64 |

Table 15: Frozen quality-only-selected image attacks on 64 reserved final seeds and fitting-holdout classes. Each attack is paired with its own supplied marked checkpoint. The distinct selected anchor weights reflect the prespecified selection policy; they are not matched-coefficient comparisons. The hybrid client returns the marked source on a quality failure; these are gated outputs, not a single adapted checkpoint. Class identities are reused with fresh generation seeds.

The quality-fallback client achieves 20/64 joint removals for the released model, 21/64 for the fixed-reference model and 20/64 for the adapted-reference model. All 64 returned images pass preservation in each case. However, this client retains the supplied generator and returns its source image whenever the raw attack fails the gates. Before fallback, raw preservation passes are only 21/64, 23/64 and 21/64, and raw removals are 58/64, 56/64 and 57/64. Consequently, its preservation rate is a property of the two-path client, not evidence that the adapted checkpoint alone preserves every source image. The clean null has 0/64 detections, which remains too small for precise low-FPR validation.

Neither training variant reduces quality-preserving removal relative to the released model. The longer adaptation horizon therefore does not rescue this meta-distillation hypothesis. Following the rejection criterion frozen before training completion, this direction stops here and is not expanded

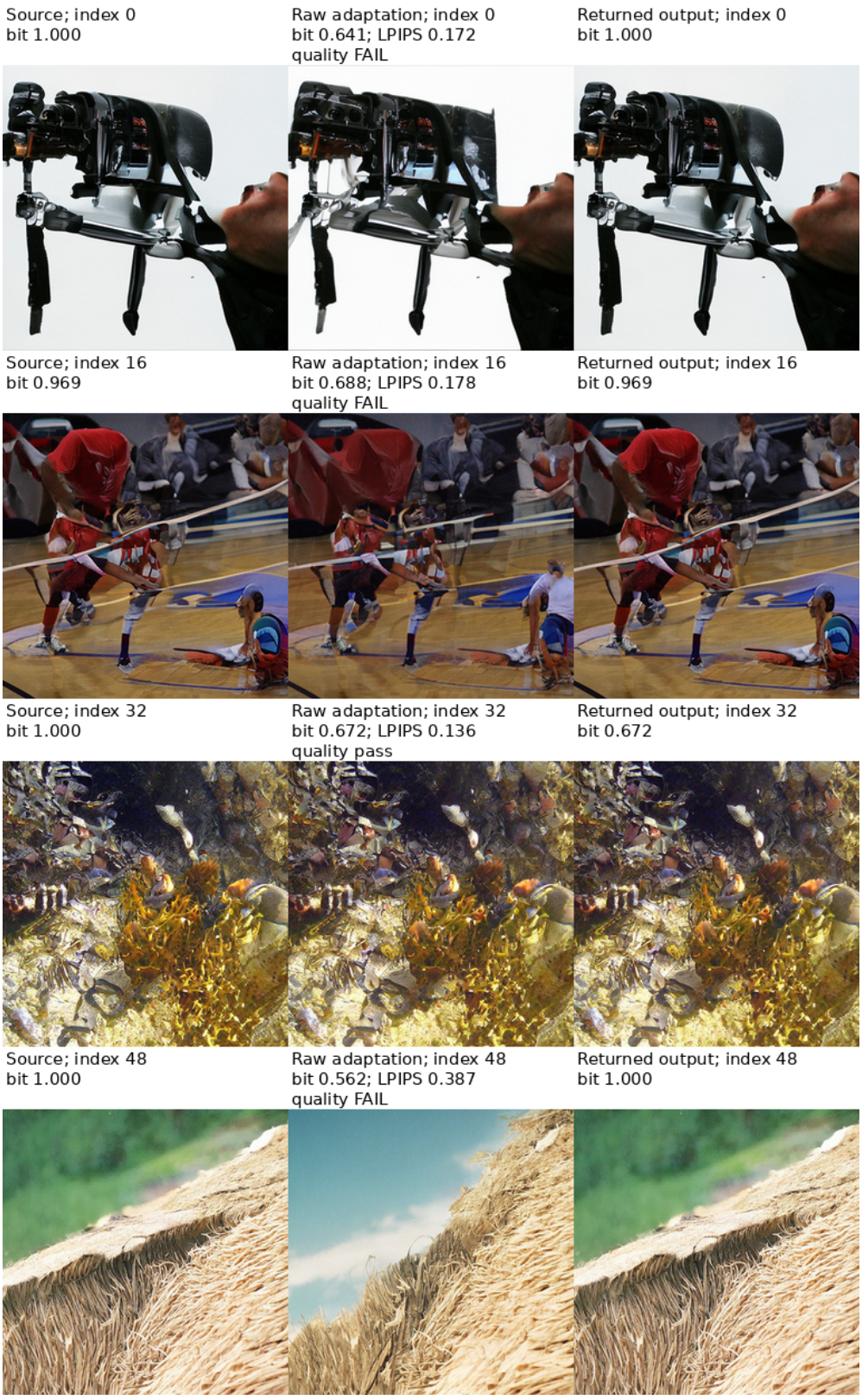



Figure 3: Released model: source, raw adaptation and returned output. LPIPS and the raw preservation decision are shown in the middle column. A low raw bit accuracy is insufficient for joint success if the content changes too much.

to additional watermark families. A different image defense would require a distinct mechanism, a source-grounded novelty assessment and a new experiment. These negative results do not support a generally robust image watermark.

The counts above retain the original frozen quality rule. Section 26 reports the later qualitative audit and revised post-hoc counts without overwriting this original result.

# 26 Qualitative recalibration and the role of SSIM

**The intended preservation target.** The revised target retains the same main objects and scene while allowing minor appearance changes. It does not require pixel identity, but excludes missing main objects, added important people, replacement of objects and major scene changes. We inspect all 32 validation pairs from the original model's unregularized 500-update attack, omitting metric values and owner decisions from the displayed panels. Six comparisons receive native-resolution closeups. Under the clarified target, assistant visual inspection assigns 19 acceptable, 11 unacceptable and two ambiguous judgements. These are model-assisted judgements rather than independent human annotations.

**The first numerical revision.** A small grid varies LPIPS limits in $\{0.2, 0.25, 0.3, 0.35\}$, SSIM minima in $\{0.5, 0.6\}$, image-image CLIP minima in $\{0.85, 0.9, 0.95\}$ and class-retention minima in $\{0.8, 0.85, 0.9, 0.95\}$. Balanced classification error on the 30 unambiguous judgements selects a class-retention minimum of $0.90$, rather than $0.95$, leaving the other three bounds unchanged. Ties favor fewer unacceptable edits admitted and then the smallest change from the old rule. On these fitting images, the revised rule accepts 13/19 judged acceptable edits, versus 11/19 previously, while excluding all 11 judged unacceptable. It still rejects six acceptable edits. This small in-sample calibration does not validate a general perceptual criterion. The initial interpretation, clarified user preference, ambiguous cases and grid results remain in the calibration receipts.

**Why an absolute worst case cannot justify removing every check.** Among watermark-evading outputs of the latest original-model attack, the worst LPIPS case loses much of a clock's casing and gains large artifacts. Another case is worst in SSIM, image CLIP and class retention; it loses an ornate foreground structure and changes the scene. These are meaningful content failures. At the other extreme, a fruit-market scene fails only SSIM, with $0.59697 < 0.6$, despite retaining its main objects and framing. A threshold boundary is not itself evidence of unacceptable content change.

**Inspecting the incremental SSIM exclusions.** To examine SSIM specifically, we retain the other three checks and inspect *all nine* outputs excluded only by SSIM across both full-training studies and all three supplied models. Selection does not depend on watermark detection. The lowest SSIM is $0.56495$; its broken-tile scene retains the same main objects despite changes to local geometry and texture. The other eight pairs also retain their main scene under the stated tolerance, including ceiling artwork, timber and fruit displays. This informal visual review motivates removing SSIM from the mandatory conjunction, while continuing to report it. It does not establish that SSIM is redundant for other data or that every local detail is preserved. Because these are already evaluated final outputs, this additional decision and all resulting comparisons are explicitly post-hoc.

The current diagnostic requires finite metrics, LPIPS at most $0.2$, image-image CLIP at least $0.9$ and class-alignment retention at least $0.9$. The owner threshold remains strict bit accuracy $> 0.6875$. Source and raw PNG hashes are checked against the final artifact freezes. Before changing any quality decision, the analysis exactly reconstructs every old raw and returned outcome for all six model/study combinations. No generator is retrained, no owner threshold is fitted and no old result is overwritten.

For the 500-inner-step revision, quality-preserving removals are now 29/64 for the released model and 28/64 for each trained variant. Raw preservation passes are 30/64 for all three. The original frozen rule gave 20/64, 21/64 and 20/64 joint removals; the intermediate class-only revision gave 27/64, 25/64

| Horizon | Model | Old joint | Revised joint | Raw quality |
|---|---|---|---|---|
| 100 | original | 20/64 | 28/64 | 33/64 |
| 100 | fixed | 22/64 | 26/64 | 31/64 |
| 100 | adapted | 22/64 | 31/64 | 35/64 |
| 500 | original | 20/64 | 29/64 | 30/64 |
| 500 | fixed | 21/64 | 28/64 | 30/64 |
| 500 | adapted | 20/64 | 28/64 | 30/64 |

Table 16: Post-hoc quality-rule reanalysis of frozen attacks. Joint removal requires loss of owner detection and preservation. Raw quality counts exclude source fallbacks. The two horizons use different generation seeds, so compare each variant to its own original baseline. This is not fresh confirmation.

and 25/64. The client can still return its marked source when the raw edit fails the remaining gates, so its raw generator does not preserve every source image. Both variants improve the removal-rate point estimate by only $1.5625$ percentage points relative to the original. Descriptive paired 95% bootstrap intervals are $[-7.81, 4.69]$ points for fixed reference and $[-6.25, 3.13]$ for adapted reference. Both include zero: this is not a confirmed defense gain.

The 100-inner-step study gives 28/64, 26/64 and 31/64 joint removals under the same current rule. Its generation seeds differ from the 500-step study, so each variant must be compared to its own original-model baseline. A separate quality-selected LoRA attack removed 0/64 watermarks from each model, including the original. The original, fixed and adapted arms preserved 64/64, 60/64 and 59/64 images, respectively. With no baseline removals, this pilot cannot distinguish defense strength. Its compact receipt remains in `tex/provenance/discarded_lora_run.json`. All revised comparisons reuse previously evaluated images and are not fresh confirmation or independent human-quality validation.

The other eight SSIM-only pairs are in `tex/figures/ssim-only-other-eight.png`.

## 27 Which next-token probabilities does the API reveal?

Let $\mathcal{V}$ be the vocabulary of size $V \in \mathbb{N}_{>0}$, $h \in \mathcal{V}^*$ a token prefix, and $k \in \mathcal{K}$ a fixed owner key in the scheme's key space $\mathcal{K}$. The vector $p(\cdot \mid h) \in \Delta(\mathcal{V})$ gives the underlying model probabilities after sampling filters. For next token $Y \in \mathcal{V}$, the key-conditional sampler law is

$$q_k(v \mid h) = \Pr(Y = v \mid h, k) \in [0, 1], \qquad q_k(\cdot \mid h) \in \Delta(\mathcal{V}).$$

The probability averages over the sampler's fresh noise at the fixed key. In the ideal fresh-context construction, averaging over independent keyed randomness restores the underlying marginal, whereas the fixed-key law carries the signal.

A client sampling exactly from $q_k(\cdot \mid h)$ at each prefix reproduces the keyed token law for that prompt. For example, the underlying model might offer two tokens with equal probability while a particular key favors one. Returning the underlying probabilities lets a client undo that preference through resampling. Returning the conditional probabilities retains it under exact resampling. Rewriting a marked answer adds another dependency because the answer itself appears in the prompt and can carry key evidence.

The implemented interface estimates each token's conditional winning probability in the clock race. Rather than count winners alone, it averages each token's chance of winning conditional on simulated competitor clocks. The raw estimate is $\hat{q} = (\hat{q}_v)_{v \in \mathcal{V}} \in [0, 1]^V$. Given positive normalizing sum $s_q = \sum_{v \in \mathcal{V}} \hat{q}_v \in (0, \infty)$, the advertised normalized estimate is $\tilde{q} = \hat{q}/s_q \in \Delta(\mathcal{V})$. Normalization introduces finite-draw bias. The interface exposes at most ten positive-probability alternatives; truncation and client reweighting can further change the law.

For probability vectors $a, b \in \Delta(\mathcal{V})$, total variation distance is $\mathrm{TV}(a, b) = \frac{1}{2} \sum_{v \in \mathcal{V}} |a_v - b_v| \in [0, 1]$. It equals the greatest discrepancy in probability assigned to any token subset. Two audits com-

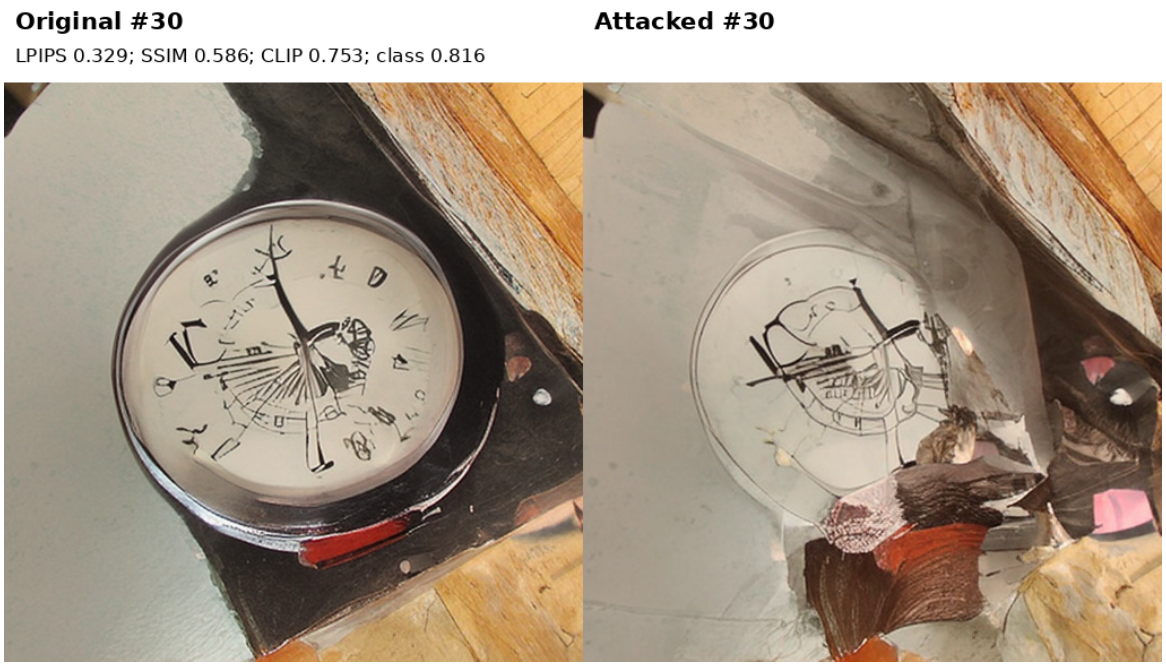



Figure 4: The worst LPIPS watermark-evading example from the latest original-model attack. The clock casing and surrounding structure change substantially. Removing all preservation checks would admit a visibly changed source.




Figure 5: The lowest SSIM among every SSIM-only rejection in both studies, irrespective of watermark outcome. Under the stated main-object and scene criterion, local texture and geometry changes are permitted. The remaining three gates pass.

pare the 256-draw estimate with high-budget race frequencies. The mean and maximum observed distances are $0.00079$ and $0.02363$ on 16 calibration prefixes, and $0.00431$ and $0.03535$ on 32 saved attack prefixes. The second set spans 1 to 170 active tokens. Native clipping and the ideal race agree on the winner in every coupled reference draw in both audits. The reference frequencies have sampling error, and these checks cover only the tested prefixes. Full tables are in `tex/generated/probability_estimation_audit.tex` and `tex/generated/probability_attack_prefix_audit.tex`.

# 28 Probability-exposure pilot: a copying control

The interface candidate in Section 27 is first evaluated on 32 newly reserved HC3 questions. Each source answer is generated by the native TextSeal2 implementation. Four blind rewriting arms compare pre-watermark versus estimated key-conditional probability exposure, each with direct resampling or confidence-dependent reference smoothing. Every arm produces three attempts per source; all 384 attempts are saved before quality evaluation or owner scoring. The quality rule requires Jina similarity at least $0.9$, bidirectional NLI entailment without truncation, exact preservation of the extracted number set, finite reference perplexity at most $1.5$ times the original, and no empty, capped or degenerate output. The first passing attempt is selected without detector feedback.

| Exposed law / attack | Preserved | Joint removals | Word copies |
|---|---|---|---|
| Pre-watermark / direct | 32/32 | 0/32 | 30/32 |
| Pre-watermark / smoothing | 32/32 | 2/32 | 25/32 |
| Conditional / direct | 32/32 | 0/32 | 26/32 |
| Conditional / smoothing | 32/32 | 0/32 | 29/32 |

Table 17: An ineffective rewriting pilot. All original answers are detected. A word copy means equality after Unicode normalization, case folding and word extraction; punctuation or whitespace changes alone therefore do not establish paraphrasing.

The selected outputs preserve almost all source token-context pairs: mean descriptive four-token-pair overlap is $0.9927$, $0.9221$, $0.9636$ and $0.9930$ in table order. These overlaps use saved generation token IDs and are not calibrated hypothesis tests. Literal string equality would misleadingly report zero exact copies because punctuation and formatting change. The observed two versus zero removals under smoothing cannot establish a meaningful defense benefit when the baseline attack mostly copies marked text. A subsequent development pilot strengthens the paraphrasing instruction and forbids source four-token completions in both target and reference supports, with explicit accounting when no allowed probability mass remains. It retains the same quality gates. This iteration reuses the 32 sources, so it is development rather than independent confirmation.

**Why a clean-oracle argument is insufficient here.** Even a pre-watermark model distribution is evaluated on a prompt containing the original marked answer. That answer depends on the secret key. Therefore the rewrite can preserve key information through its input, although the probability computation itself does not consult the key. An impossibility argument for key-independent prompts and a clean resampling oracle does not apply to this experiment without accounting for that dependence. Conversely, the exact-law identity for conditional exposure does not guarantee robustness under smoothing, truncation or substantive rewriting. Both distinctions are required when interpreting the next pilot.

## 28.1 Stronger rewriting with source-pair blocking

The completed second pilot keeps all 32 marked source answers and the same quality gates. Its stronger instruction and source four-token blocking substantially change the attack. The blocking operates after mixing target and reference supports; if no mass remains, the unmasked response is used and the

fallback is recorded. This is an explicit exception rather than a guarantee of zero copied source pairs. The native detector is applied only after all attempts and the first quality-passing selections are frozen.

| Exposed law / attack | Preserved | Raw removed | Joint removed | Word copies |
|---|---|---|---|---|
| Pre / direct | 25/32 | 32/32 | 25/32 | 0/32 |
| Pre / smoothing | 22/32 | 32/32 | 22/32 | 0/32 |
| Conditional / direct | 24/32 | 7/32 | 5/32 | 0/32 |
| Conditional / smoothing | 27/32 | 16/32 | 12/32 | 0/32 |

Table 18: Development rewrite test after blind quality-only selection; all 32 originals are detected. Raw removal includes failed-preservation outputs, while joint removal requires the unchanged similarity, entailment and fluency gates. Reused sources do not constitute independent confirmation.

This test shows a development benefit: valid removals decrease from 25 to 5 for direct resampling and from 22 to 12 for smoothing. Preservation passes change from 25 to 24 and from 22 to 27, respectively, so the lower joint removal is not explained solely by rejecting more conditional-interface outputs. Nevertheless, this is a single-key, 32-source development comparison with three attempts per arm. The source identities have already been used in the copying control. The fresh human-null diagnostic remains 0/32 at the native nominal threshold, which cannot precisely validate a $0.1\%$ false-positive rate. Independent prompts, additional keys, conditional-estimator checks and adaptive ratio attacks remain required.

## 29 Adaptive probability reweighting defeats conditional exposure

The source-blocked pilot in Section 28 motivates a stronger client rather than a robustness claim. On the same 32 development sources, an independent Qwen3.5-0.8B reference supplies probabilities on the target API's advertised top-ten token IDs. At a fixed prefix, let $\mathcal{V}$ be the vocabulary and $C \subseteq \mathcal{V}$ its nonempty advertised positive-probability support, of size $K_C = |C| \in \{1, \dots, 10\}$. For token $v \in C$, the scalar advertised probability is $q_v \in (0, 1]$ and the numerically floored reference probability is $r_v \in (0, 1]$. The vectors $(q_v)_{v \in C}, (r_v)_{v \in C} \in (0, 1]^{K_C}$ contain the advertised-support coordinates before client normalization. With scalar tilt strength $\beta \in [1, \infty)$, define weights $w_v = q_v^{1-\beta} r_v^{\beta} \in (0, \infty)$ and normalized client law

$$a_v = \frac{w_v}{\sum_{u \in C} w_u} \in (0, 1], \qquad (a_v)_{v \in C} \in \Delta(C).$$

The three attempts use $\beta \in \{1, 2, 4\}$. At $\beta = 1$, advertised probabilities cancel and the client samples the reference distribution restricted to $C$. Increasing $\beta$ favors tokens that the reference rates more highly relative to the advertised model. The implementation excludes zero-probability lanes, computes weights in log space with an explicit numerical floor, and applies the same source-pair blocking after reweighting. This is an exposed-support adaptation of prior ratio attacks, not a replication of their repeated-query protocol or an optimality result (Omidi et al., 2026).

| Exposed law / attack | Preserved | Raw removed | Joint removed | Word copies |
|---|---|---|---|---|
| Pre / ratio tilt | 27/32 | 32/32 | 27/32 | 0/32 |
| Conditional / ratio tilt | 21/32 | 32/32 | 21/32 | 0/32 |

Table 19: Development rewrite test after blind quality-only selection; all 32 originals are detected. Raw removal includes failed-preservation outputs, while joint removal requires the unchanged similarity, entailment and fluency gates. Reused sources do not constitute independent confirmation.

Every quality-selected output evades the native detector under both interfaces. Quality passes are 27/32 for the pre-watermark interface and 21/32 for conditional exposure. Consequently, the

smaller conditional joint-removal count is entirely quality attrition: it is not evidence that the watermark survives. This negative control overturns a general robustness interpretation of the earlier direct-resampling result. All attempts precede owner evaluation, and selection still uses the first attempt passing the unchanged quality gates.

The 128 fresh-source confirmation cohort was frozen before this development outcome. It evaluates all six unguarded arms and therefore tests whether both the restricted benefit and adaptive failure transfer to new questions. It does not test the later support-pruning candidate, does not supply independent keys, and must not be relabeled as confirmation of that candidate.

# 30 Fresh-source confirmation separates restricted benefit from adaptive failure

The confirmation protocol reserves 128 new HC3 question identities, excludes the 1,712 retained questions and all 96 initial development/calibration identities, and splits generation into two disjoint 64-question shards. The shards share 64 separate public calibration identities. They use the same frozen watermark key and six source-blocked clients: direct resampling, reference smoothing and surrogate-ratio reweighting under each probability interface. Every client produces three attempts per source before quality-only selection and owner evaluation. The 2,304 attempts are retained. This cohort was fixed before the ratio development result and contains no support-pruning treatment.

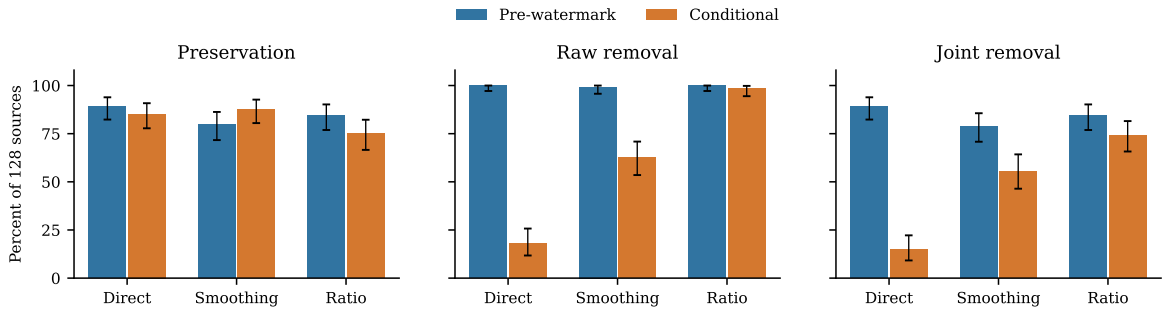



Figure 6: Three distinct outcomes for the fresh confirmation cohort. Error bars are exact binomial 95% intervals, conditional on the single key and evaluated models. Conditional exposure retains substantial signal under direct resampling, less under smoothing, and almost none under adaptive ratio reweighting. Lower joint removal alone cannot distinguish signal retention from quality attrition.

| Interface / client | Preserved | Raw removed | Joint removed | Joint 95% CI |
|---|---|---|---|---|
| Pre / direct | 114/128 | 128/128 | 114/128 | [82.3, 93.9]% |
| Pre / smooth | 102/128 | 127/128 | 101/128 | [70.8, 85.6]% |
| Conditional / direct | 109/128 | 23/128 | 19/128 | [9.2, 22.2]% |
| Conditional / smooth | 112/128 | 80/128 | 71/128 | [46.4, 64.3]% |
| Pre / ratio | 108/128 | 128/128 | 108/128 | [76.9, 90.2]% |
| Conditional / ratio | 96/128 | 126/128 | 95/128 | [65.7, 81.5]% |

Table 20: Fresh 128-source, single-key confirmation after blind first-quality-passing selection from three attempts. Initially detected sources form the removal denominator. Intervals are exact binomial intervals, conditional on the evaluated key and models; paired differences are reported separately.

All 128 originals are detected. Under direct resampling, changing exposure reduces joint removal from 114/128 to 19/128, while preservation passes change from 114/128 to 109/128. The paired difference is $-74.2$ percentage points, with a descriptive 95% paired prompt-bootstrap interval

$[-82.0, -65.6]$. Under smoothing, joint removal decreases from 101/128 to 71/128, with a paired difference of $-23.4$ points and interval $[-32.8, -14.1]$. These results confirm a restricted-client benefit on fresh questions.

The adaptive client changes the interpretation. Ratio reweighting removes 128/128 watermarks under pre-watermark exposure and 126/128 under conditional exposure. The latter preserves 96/128 outputs, of which 95 evade detection. Its 95/128 joint successes therefore reflect only one detected quality-preserving rewrite. The paired joint difference against pre-watermark exposure is $-10.2$ points, with interval $[-18.0, -2.3]$, but quality attrition explains most of that gap. A statistically distinguishable joint-success difference is not equivalent to robust watermark retention.

The intervals resample paired prompt identities 10,000 times with a fixed analysis seed. They are descriptive post-hoc uncertainty summaries, not prespecified multiple-comparison tests or uncertainty over training seeds and keys. The pooled human-null diagnostic is 0/128 at the native nominal threshold $0.001$; this small single-key set cannot validate a $0.1\%$ fixed-key false-positive rate precisely. No selected rewrite is a normalized word copy. Token-context overlap, source-blocking fallback counts and both shard-level tables remain in the analysis receipts.

# 31 Pruning advertised alternatives

A client can reweight the advertised tokens, including those carrying weak owner evidence. Support pruning removes some alternatives while retaining most of their estimated probability mass.

**The optimization at one prefix.** Let $\mathcal{V}$ be the vocabulary of size $V \in \mathbb{N}_{>0}$, $h \in \mathcal{V}^*$ a prefix, and $k \in \mathcal{K}$ an owner key in the scheme's key space $\mathcal{K}$. At that prefix, $\tilde{q} = (\tilde{q}_v)_{v\in\mathcal{V}} \in \Delta(\mathcal{V})$ is the normalized estimate of the keyed sampler law. The ordinary advertised support is a nonempty set $C_0 \subseteq \mathcal{V}$ of at most ten positive-probability tokens. The native owner evidence function is $e_k : \mathcal{V}^* \times \mathcal{V} \to [0, \infty)$. For token $v \in C_0$, it returns scalar $e_k(h, v) \in [0, \infty)$. Larger evidence favors detection. Given scalar mass-loss budget $\delta \in [0, 1)$, choose a nonempty subset $S \subseteq C_0$ solving

$$\max_{\emptyset\neq S\subseteq C_0} \min_{v\in S} e_k(h, v) \quad \text{subject to} \quad \sum_{v\in S} \tilde{q}_v \geq (1-\delta) \sum_{v\in C_0} \tilde{q}_v.$$

The objective raises the weakest evidence a supported client can choose. Sort tokens by decreasing evidence and accumulate probability mass until the constraint holds, retaining ties at the cutoff. For probabilities $(.80, .15, .05) \in \Delta(\{1, 2, 3\})$ and evidence $(4, 3, .1) \in \mathbb{R}^3$, dropping the last token raises the minimum evidence from .1 to 3 while discarding only $5\%$ of mass.

**What the budget controls.** Let $b_0 = \sum_{v\in C_0} \tilde{q}_v \in (0, 1]$ and $b_S = \sum_{v\in S} \tilde{q}_v \in (0, b_0]$. Define the renormalized probability vectors $P_0, P_S \in \Delta(\mathcal{V})$ by

$$P_0(v) = \frac{\tilde{q}_v \mathbf{1}\{v \in C_0\}}{b_0}, \qquad P_S(v) = \frac{\tilde{q}_v \mathbf{1}\{v \in S\}}{b_S}.$$

Their scalar total variation distance is exactly $\mathrm{TV}(P_0, P_S) = 1 - b_S/b_0 \leq \delta$. This bounds how much pruning changes the advertised distribution at this prefix. Sequential changes can accumulate, so answer preservation still needs direct evaluation.

For an output $y = (y_1, \ldots, y_L) \in \mathcal{V}^L$ of length $L \in \mathbb{N}_{>0}$, let $\mathcal{T} \subseteq \{1, \ldots, L\}$ contain the positions scored by the native first-occurrence rule. At each such position, the prefix is $h_t \in \mathcal{V}^*$, the retained support is nonempty $S_t \subseteq \mathcal{V}$, and its evidence floor is $a_t = \min_{v\in S_t} e_k(h_t, v) \in [0, \infty)$. If every scored token $y_t$ belongs to $S_t$, then

$$S_{\text{path}} = \sum_{t\in\mathcal{T}} e_k(h_t, y_t) \geq \sum_{t\in\mathcal{T}} a_t,$$

where $S_{\text{path}} \in [0, \infty)$ is the summed native evidence. A bound for saved token IDs applies to decoded text only when the scored path is preserved by retokenization. Reference smoothing and independent rewriting may choose tokens outside the retained supports.

## 32 Support pruning resists a supported ratio client in development

**A matched development comparison.** The experiment in Section 31 has completed generation, preservation selection and native owner scoring. It reuses the same 32 marked answers as the earlier blocked-source and ratio experiments. The source-answer files are byte-identical; the watermark key, sampling settings and quality-selection rule also match. Each interface uses its own public-prefix confidence calibration. The guard was designed after observing the earlier ratio failure, so this is a development comparison rather than an independent confirmation. Three attempts per source are frozen before selecting the first quality-passing output. No source is replaced by the original answer when all attempts fail.

| Exposed law / attack | Preserved | Raw removed | Joint removed | Word copies |
|---|---|---|---|---|
| Prune 10 / direct | 23/32 | 4/32 | 3/32 | 0/32 |
| Prune 10 / smooth | 25/32 | 18/32 | 15/32 | 0/32 |
| Prune 10 / ratio | 25/32 | 4/32 | 3/32 | 0/32 |
| Prune 20 / direct | 23/32 | 3/32 | 3/32 | 0/32 |
| Prune 20 / smooth | 27/32 | 15/32 | 13/32 | 0/32 |
| Prune 20 / ratio | 24/32 | 2/32 | 1/32 | 0/32 |
| Independent reference | 22/32 | 32/32 | 22/32 | 0/32 |

Table 21: Blind quality-only selection on 32 sources; 32 originals are detected. Raw removal includes failed-preservation outputs, while joint removal requires the unchanged similarity, entailment and fluency gates. Reused development sources do not constitute independent confirmation.

**A gain against supported reweighting.** Without pruning, conditional ratio reweighting removes all 32 selected watermarks, including all 21 quality-preserving rewrites. Retaining at least 90% of the ordinary top-ten estimated mass reduces raw removals to 4/32 and joint removals to 3/32, with 25/32 quality passes. At 80% retained mass, the corresponding counts are 2/32 raw removals, 1/32 joint removals and 24/32 quality passes. Thus the development gain is not explained by rejecting more outputs. Among quality-preserving outputs, watermark removal falls from 21/21 to 3/25 and 1/24. All selected guarded outputs differ from their sources under normalized-word equality; this check excludes literal copying but does not certify substantive rewriting or factual equivalence.

**The access boundary is visible in the failures.** Direct conditional resampling has 5/32 joint removals without pruning and 3/32 at either guard budget, a small development difference. Smoothing does not improve: its 12/32 baseline joint removals become 15/32 and 13/32. The independent reference removes all 32 watermarks, including all 22 outputs passing the preservation gates. These controls show that supported ratio resistance does not establish resistance to external rewriting. They also prevent describing the guard as a generally robust watermark or a defense for open-weight models.

**Checking the mathematical relation against native scoring.** All 672 attempts were scored as actual decoded text and as saved generated token IDs. Every generated-ID statistic satisfies the frozen pathwise evidence bound, including the native first-occurrence masking rule. Decode–encode roundtrips hold for all 96 attempts in each direct and ratio arm, 96/96 and 94/96 in the two smoothing arms, and 92/96 independent-reference attempts. The remaining paths receive no decoded-text certificate. At the native nominal threshold, the bound certifies 40/96 ratio paths at 90% retention and 62/96 at 80%

retention. These are certificates across all attempts, not quality-selected success counts; an uncertified path can still be detected. The relation does not prove the nominal Gamma calibration for adaptive paths. The separate human-null audit addresses one fixed-key diagnostic, rather than repairing this limitation.

**What remains to establish.** The existing 128-question confirmation concerns conditional exposure without support pruning. It cannot confirm this new treatment. Independent prompts, keys and model pairs, an attack that adapts to the guard, and additional watermark families are still required. The complete native scores, quality outcomes, support-path certificates and selected strings are retained in the hashed owner results and portable development receipts.

## 33 A larger independent human-null audit

The 128-source confirmation contains only 128 human-null prefixes, too few to assess a nominal $0.1\%$ error rate precisely. We therefore freeze a separate 4,096-prefix diagnostic before computing its owner scores. It uses HC3 human-labelled answers from question-hash buckets 65–89, outside the probability-interface evaluation and public-calibration buckets. All retained, development, confirmation and calibration question identities are excluded. Questions and first-100-word human prefixes are deduplicated, and known human-prefix hashes are excluded. No threshold or key is fitted on this set.

The native TextSeal2 v2 decision remains $p \leq 0.001$, with null-tail $p$-value $p \in [0, 1]$. It flags 5 of 4,096 prefixes: an empirical false-positive rate of $0.122\%$, with an exact binomial 95% interval $[0.040\%, 0.285\%]$. Every prefix has at least one scored context/token pair. This cohort is compatible with the nominal rate but does not precisely establish that rate for every key, dataset or adaptively selected input. The interval is descriptive under the usual independent-sample model, conditional on the existing key and this human-labelled population.

The audit runs entirely on CPU. It calls the released native copula-score primitive in batches, preserves the native first-occurrence context/token-pair rule, and uses the unchanged Gamma-tail function. Before computing the audit, it checks agreement with the scalar detector on 12 probes: empty, short and repeated-token cases plus eight fixed-position real prefixes. All scored-pair counts and per-token values agree to the declared $10^{-12}$ tolerance. The frozen manifest, individual $p$-values, source hash and result hashes remain in the audit receipt. This is an implementation and null-behavior check, not evidence of attack resistance or a new calibration procedure.

# Part V
# Synthesis and reproducibility

## 34 What the evidence supports

The experiments show adaptation against fixed detectors and substantial suppression of several measured watermarks. They also expose limits of raw evasion: BIRA suppression accompanies lower bidirectional entailment; some DC-AE weight merges suppress detection while damaging class alignment; and probability resampling can bypass the sampler under a stronger oracle than a closed provider normally reveals.

None of these observations establishes universal watermark erasure. A negative result from one owner detector does not exclude an unknown second mark. Similarity metrics do not certify semantic equivalence or image usefulness. Preserved class alignment does not certify diversity, realism or absence of memorization.

# 35 Reproducibility and evidence

The repository README gives the build and experiment commands. The detector catalogue and experimental records in `tex/records/` retain protocols, full tables and historical comparisons. The migration map in `tex/provenance/migration.json` identifies their original sources. The watermarking results index in `docs/watermarking/README.md` provides the fixed-sample galleries and complete example texts. Full recent rewriting examples are in `tex/generated/` and the corresponding sample JSON files are in `tex/provenance/`.

The latest experiment receipts in `tex/provenance/` include the frozen selections and owner scores.

- Conditional exposure: `text_exposure_confirmation_results.json`
- Support pruning: `text_exposure_guard_results.json`
- Human-null audit: `textseal2_human_null_audit.json`
- Image preservation reanalysis: `revised_quality_no_ssim_results.json`

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